Applied Study

Supplementary material for the CountBCF manuscript

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1 Applied Study

The simulation experiments establish the operating characteristics of CountBCF under controlled data-generating mechanisms. We now assess it in a setting where the outcome, exposure, and scientific benchmarks are all real. We analyse solid-cancer incidence in the Radiation Effects Research Foundation (RERF) Life Span Study (LSS) of survivors of the Hiroshima and Nagasaki atomic bombings. The LSS is the principal source of quantitative evidence on radiation-related cancer risk in humans, with detailed follow-up and well-characterised dosimetry supporting successive incidence analyses [Thompson et al., 1994, Preston et al., 2007, Grant et al., 2017].

The application has two purposes: to compare CountBCF's average effects with conventional risk models, and to assess whether it recovers established variation with sex, age at exposure, and attained age without specifying those interactions. The all-solid-cancer analysis recovers the principal average and age-related patterns. Exploratory city differences at low dose are sensitive to the reference population. Reference-group composition accounts for Hiroshima's estimated deficit and part of the city contrast, while a modest Nagasaki excess remains unexplained.

This supplementary version reports the complete tables, figures, sensitivity analyses, and diagnostics; a condensed account appears in the main text. All analyses use the public 1958–1998 person-year table. Code, analysis outputs, and environment information accompany the supplementary materials.

1.1 Data validation against the published analysis

Before any modelling, the pipeline verifies that our handling of the public file reproduces the published totals. Table 1 compares our analysis frame with Table 4 of Preston et al. [2007]. Every case count and every head count matches exactly, and person-years agree to a maximum relative difference of 2.6×10^{-4}, which is attributable to rounding in the published table. Matching these totals supports the recorded cohort construction, but cannot exclude all data-processing discrepancies; model specification and analytic restrictions may also produce later differences.

The composition of the all-solid endpoint also required verification, because it determines which site-specific mechanisms can drive the aggregate result. The all-solid total decomposes cell by cell, with zero discrepancies, as the sum of oral cavity, digestive, respiratory, skin/bone/connective

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Table 1: Validation of the analysis frame against Table 4 of Preston et al. [2007]. Case and person counts match exactly; the person-year differences reflect rounding in the published table.

Stratum	Cases	Published	Person-years	Published	People	Published
Not in city	3,994	3,994	680,744	680,744	25,247	25,247
$<$ 0.005 Gy	5,603	5,603	918,200	918,199	35,545	35,545
0.005–0.1	4,406	4,406	729,604	729,603	27,789	27,789
0.1–0.2	968	968	145,925	145,925	5,527	5,527
0.2–0.5	1,144	1,144	153,886	153,846	5,935	5,935
0.5–1	688	688	81,251	81,251	3,173	3,173
1–2	460	460	41,413	41,412	1,647	1,647
2–4	185	185	13,712	13,711	564	564
Male	7,969	7,969	1,040,281	1,040,278	42,902	42,902
Female	9,479	9,479	1,724,454	1,724,452	62,525	62,525
Hiroshima	12,931	12,931	1,967,602	1,967,601	73,430	73,430
Nagasaki	4,517	4,517	797,133	797,129	31,997	31,997
Total	17,448	17,448	2,764,735	2,764,730	105,427	105,427

tissue, breast, female genital, male genital, urinary, renal pelvis and ureter, central nervous system, thyroid, other solid, and thymus. Two codebook features make this non-obvious: the urinary group (661 known-dose cases) omits renal pelvis and ureter (80 cases) even though the documentation places the latter under the urinary system, and thymus (12 cases) sits outside the respiratory group. Restoring both recovers 17,448 exactly. Group by group against Table 2 of Preston et al. [2007], oral cavity (277), digestive (10,052), respiratory (2,001), skin, bone and connective tissue (400), breast (1,082), female genital (1,457), male genital (420), central nervous system (281), thyroid (471), stomach (4,730), lung (1,759), colon (1,516) and liver (1,494) all match the published counts to the case. Within the skin, bone and connective tissue group, non-melanoma skin cancer contributes 330 cases and melanoma 17, again exactly reproducing the published figures. Non-melanoma skin cancer is therefore inside the all-solid aggregate, consistent with the published description of 17,448 first primary cancers including non-melanoma skin cancer. This matters because non-melanoma skin is one of the few sites whose ERR varies significantly with age at exposure in the published heterogeneity tests, making it a candidate driver of the aggregate's age-at-exposure shape. Section 1.9 tests that candidate directly.

1.2 Data, treatment design, and estimands

The source is the public 1958–1998 LSS solid-cancer incidence person-year table. Each row is a stratum defined by city, sex, age at exposure, attained age, follow-up period, colon-dose category, and distance from the hypocentre (with a separate not-in-city category); it records cancer cases and person-years at risk. The analytic endpoint is all first primary solid cancers. Its 17,448 cases among 105,427 cohort members reproduce the published totals in Preston et al. [2007], as documented in Section 1.1.

Unknown-dose records are excluded: 1,236 cells carrying 174,631 person-years and 1,197 cases. The primary analysis is restricted to in-city survivors and uses the standard DS02 weighted colon dose, gamma dose plus ten times neutron dose, expressed in Gy. We contrast a common reference group below 5 mGy against three exposure groups: low (5–$<$ 100 mGy), medium (100–$<$ 500 mGy), and high (500–$<$ 2000 mGy). Survivors at or above 2 Gy are excluded from the primary contrasts and restored in a sensitivity fit. Each group is fitted as a binary CountBCF analysis against the

same reference group. This categorical construction makes the causal contrast explicit and does not impose a dose-response curve. One point of nomenclature deserves mention: the dose columns are weighted, dose-error-adjusted DS02 organ doses in Gy despite the codebook labelling them mGy, whereas the dose-category cutpoints are in mGy.

Survivors not in either city are excluded from the primary contrast because their probability of receiving an in-city bomb dose is zero, so including them in the reference group of a exposed-versus-reference contrast would violate the overlap condition that CountBCF relies on. This is not an objection to their use in a regression that borrows their follow-up to characterize the baseline rates, which is the conventional LSS treatment. We therefore include them in a sensitivity analysis with a not-in-city indicator in the prognostic forest, the CountBCF analogue of the standard baseline adjustment, and compare the two fits in Section 1.9. The published analysis provides the benchmark for this check, reporting that dose parameters are virtually unchanged when the not-in-city group is excluded and that including it without baseline adjustment inflates the ratio-of-rates ERR by about 8%.

The prognostic forest receives city, sex, age at exposure, attained age, and the estimated propensity score; the moderating forest receives the same substantive covariates. Calendar year, birth cohort, and time since exposure are deterministic functions of the two age variables and are not entered separately. The relation is year = age − age at exposure + t_0, where t_0 is the date of the detonation in decimal years. Across all 26,806 cells the implied t_0 lies between 1945.594 and 1945.611, with a median of 1945.597 in Hiroshima and 1945.605 in Nagasaki, which are 6 and 9 August 1945. The relation is thus exact to within about six days and is a property of the calendar rather than of the coding. City is included in the moderator set. The Preston et al. specification refitted here includes city in the baseline only; conventional LSS analyses can also test city as an ERR modifier [Preston et al., 2007, Grant et al., 2017]. The propensity score is the estimated probability of membership in the exposed dose group given the measured covariates. Ground distance is omitted from the primary specification because it nearly determines dose, leaving little overlap when conditioned on. This choice restricts adjustment: distance-related differences in background risk remain potential sources of confounding. Sections 1.10, 1.11, and 1.15 examine this limitation.

Section 1.10 therefore adds a separate family of fits in which the coarse $< 3/\geq 3$ km ring, which is a baseline-composition variable rather than the continuous assignment variable, does enter both the score and the prognostic forest. The propensity-score adjustment follows the regularization-induced-confounding strategy of Hahn et al. [2020], Rosenbaum and Rubin [1983].

With person-years as an offset, $\exp(\mu_{f,i})$ is the reference-arm rate in cell i and $\tau_{f,i}$ is the log rate ratio. We report both an absolute and a relative conditional effect:

$$\mathrm{EAR}_i = \{\exp(\mu_{f,i} + \tau_{f,i}) - \exp(\mu_{f,i})\} \times 10^4, \qquad \mathrm{ERR}_i = \exp(\tau_{f,i}) - 1. \tag{1}$$

The first is an excess absolute rate in cases per 10^4 person-years; the second is an excess relative rate (ERR). For a specified averaging population G, we also report a dose-group ERR,

$$\mathrm{ERR}_G = \frac{\sum_{i \in G} w_i r_{1,i}}{\sum_{i \in G} w_i r_{0,i}} - 1, \qquad w_i = \text{person-years in cell } i,$$

where $r_{1,i}$ and $r_{0,i}$ are the counterfactual rates under membership in the exposed and reference dose groups. The *cohort-averaged ERR* uses all cells in the two-group comparison; the *exposed-group ERR* uses only cells in the exposed group. The conventional causal labels ATE (average treatment effect) and ATT (average treatment effect among the treated) identify these averaging populations. Here "arm" means one of the two dose groups, not a randomized assignment. Subgroup ERRs use the same ratio of rates within the stated subgroup.

A dose-group ERR differs from the person-year-weighted average of the cell ERRs. For high-dose all-solid cancer, these give rate ratios of approximately 1.49 and 1.84, respectively, because high cell ERRs tend to occur where baseline rates are low. We compare ratios of averaged rates with parametric g-computation summaries and report the cell-average ERR separately [Hernán and Robins, 2006, Robins, 1986]. Every credible interval and subgroup difference is calculated from posterior draws. A dose-group ERR is a categorical contrast, not an ERR per Gy and not an estimate evaluated at a fixed age profile.

The causal interpretation is conditional on consistency, no interference, positivity, and conditional exchangeability given the available covariates. Dose resulted from location and shielding at the time of the detonations, not a behavioural treatment choice, but the study remains observational. In particular, the public table contains no individual smoking, dietary, or infection information. The estimates should therefore be read as causal under these assumptions and as adjusted associations otherwise.

1.3 Descriptive structure of the analysis table

Table 2: Descriptive summary of the all-solid-cancer analysis table. The reference and three treatment groups are in-city groups. Rates are crude and reported before covariate adjustment. The final column is the marginal variance-to-mean ratio of the cell counts.

Dose group	Cells	Subjects	Person-years	Cases	Rate/10^4 PY	Mean dose (Gy)	Zero cells (%)	Var/mean
Not in city	1,285	25,247	680,744	3,994	58.7	0	44.1	8.10
Reference, 0–< 5 mGy	2,568	35,545	918,200	5,603	61.0	0.00096	47.2	5.37
Low, 5–< 100 mGy	5,973	27,789	729,604	4,406	60.4	0.0308	68.6	3.42
Medium, 100–< 500 mGy	7,598	11,462	299,812	2,112	70.4	0.2354	80.8	1.65
High, 500–< 2000 mGy	5,681	4,820	122,664	1,148	93.6	0.9216	83.9	1.34

Table 2 illustrates why adjustment is necessary. The low-dose crude rate of 60.4 per 10^4 person-years is slightly below the reference rate of 61.0, although the adjusted dose response rises across the medium and high-dose groups. These crude comparisons do not account for differences in covariate composition. Age at exposure, attained age, and sex are almost perfectly balanced across groups, with standardized mean differences between 0.002 and 0.047, whereas city is not: Hiroshima accounts for 58% of the reference group and 78% of the low-dose group, giving standardized mean differences between 0.19 and 0.56. City is therefore the one covariate on which the propensity model has real work to do, and it is why the adjusted analyses move the low-dose-group estimate.

The cells are sparse but not, on inspection, genuinely overdispersed. Roughly 74% of the analysis cells are zero and the marginal variance-to-mean ratio runs from 1.3 to 5.4. Once a flexible baseline is conditioned on, consisting of sex-specific splines in log attained age and in age at exposure plus a city term, the extra-Poisson variation essentially disappears: the fitted negative-binomial size is $\theta \approx 100$, implying a variance-to-mean ratio of 1.006 at the mean fitted rate; the negative-binomial AIC is no better than the Poisson AIC; and the likelihood-ratio test for the negative binomial over the Poisson gives $p = 0.5$. The quasi-Poisson scale, at a Pearson χ^2/df of 1.69, is inflated mostly by the sparse-cell behaviour of the Pearson statistic, since the deviance per degree of freedom is 0.59. The apparent overdispersion in this table is thus baseline heterogeneity rather than extra-Poisson noise, and a model flexible enough in the baseline absorbs it. We nevertheless run CountBCF with the negative-binomial likelihood because it nests the Poisson and lets the data decide; its dispersion posterior is reported in Section 1.5. Many zero cells are expected in thin person-year strata; their frequency alone does not distinguish sampling zeros from structural zeros.

The crude data already display the canonical gradient, which is the pattern against which any model output must be read. Comparing the high-dose group to the reference within age-at-exposure strata, the crude rate ratio runs 3.41, 1.90, 1.51, 1.35, and 1.41 across age-at-exposure groups 0–9, 10–19, 20–29, 30–39, and 40 or more for women, and 1.71, 1.72, 1.41, 1.12, and 1.20 for men.

1.4 Parametric anchors

Three parametric anchors are fitted on the identical in-city analytic set: a refit of the canonical excess-relative-risk and excess-absolute-risk models, a fully categorical dose response, and adjusted generalised linear models for the marginal contrasts.

1.4.1 Refitting the canonical risk model

The standard model is refitted by maximum Poisson likelihood on the same in-city analysis set, using a sex-specific spline baseline in log attained age and in age at exposure plus a city term, with

$$\mathrm{ERR}(d, e, a, s) = \beta_s \, d \, \exp\{\alpha_e (e - 30) + \alpha_a \log(a/70)\}\,, \tag{2}$$

and the additive analogue for the EAR, where d is weighted colon dose, e is age at exposure, a is attained age, and s is sex. Because this refit is the comparator against which the CountBCF surface is measured in Section 1.7, it matters that it reproduces the published fit. Table 3 shows that every one of the ten published parameters is recovered inside its published 90% confidence interval. The parametric yardstick is therefore the established risk model itself, rather than a simplified regression surrogate.

Table 3: Refit of the canonical ERR and EAR risk models on the in-city analytic set, compared with Table 10 of Preston et al. [2007]. Intervals are 90% confidence intervals. Every refitted parameter falls inside the published interval. The published fit includes not-in-city survivors, whereas this refit is restricted to in-city survivors, which also contributes to numerical differences.

Parameter	Refit (90% CI)	Published (90% CI)
ERR per Gy, male, attained age 70, exposed at 30	0.336 (0.245, 0.426)	0.35 (0.28, 0.43)
ERR per Gy, female, attained age 70, exposed at 30	0.554 (0.438, 0.670)	0.58 (0.43, 0.69)
ERR female-to-male ratio	1.65	1.6 (1.31, 2.09)
ERR change per decade of age at exposure	−15.2% (−25.6%, −3.1%)	−17% (−25%, −7%)
ERR attained-age power	−1.87 (−2.45, −1.28)	−1.65 (−2.1, −1.2)
EAR per 10^4 PY-Gy, male	38.9 (25.8, 52.1)	43 (33, 55)
EAR per 10^4 PY-Gy, female	57.8 (47.6, 68.1)	60 (51, 69)
EAR female-to-male ratio	1.49	1.4 (1.10, 1.79)
EAR change per decade of age at exposure	−23.5%	−24% (−32%, −16%)
EAR attained-age power	+2.25 (1.73, 2.77)	2.38 (1.9, 2.8)

Fitting one parameter per dose category on the same baseline gives 18 category-specific estimates without effect modification. The blue line in Figure 1 is an ordinary least-squares fit through these point estimates and the origin, shown as a descriptive trend. It assigns each category equal weight and does not account for the covariance among the category estimates. The published 0.47 per Gy instead comes from a model with effect modification standardized to attained age 70 and exposure age 30 [Preston et al., 2007]; it is not the coefficient of this descriptive line. The figure also shows the exposed-group CountBCF ERRs, with high-dose value 0.487 (0.386, 0.580), the same categorical contrast reported below.

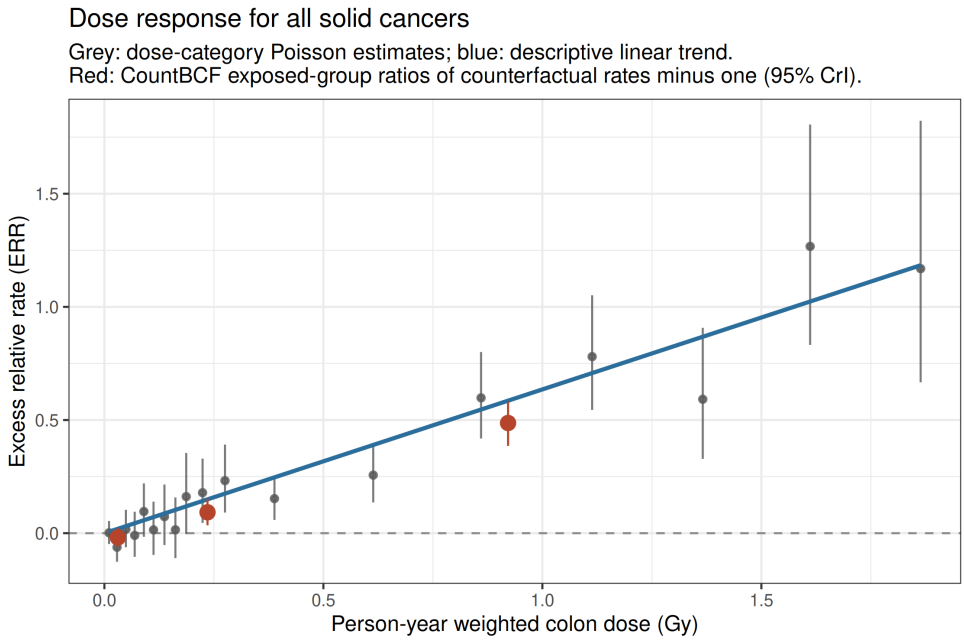


Figure 1: Dose-category ERRs from the Poisson model without effect modification and a descriptive line through their point estimates, compared with exposed-group CountBCF ratios of person-year-weighted counterfactual rates minus one. Horizontal positions are person-year-weighted mean colon doses in the exposed categories. CountBCF intervals are 95% credible intervals; parametric intervals are 95% confidence intervals.

1.4.2 Generalised linear model marginal effects

Table 4 reports g-computation marginal effects from adjusted Poisson and negative-binomial models. The Poisson and negative-binomial fits are nearly indistinguishable for all-solid cancer, consistent with the dispersion diagnostics of Section 1.3. Adding dose by modifier interactions improves the all-solid negative-binomial fit by $\Delta\text{AIC} = 54$ (26,931 against 26,985), so heterogeneity is detectable parametrically once the analyst knows to look for it.

Two features of the parametric family are worth carrying forward. The first is the improvement just noted, which establishes that the heterogeneity CountBCF reports in Section 1.6 is present in the data and not a property of the forest. The second is that the unadjusted models are unusable: a minimal negative-binomial model for high-dose stomach returns a rate ratio of 63 with a 95% interval of (0.08, 521), and a minimal negative-binomial model for low-dose all-solid cancer flips to a rate ratio of 0.90 against the Poisson's 0.99. These minimal fits illustrate the need to model background rates and to check numerical stability. They do not show that flexible parametric analyses are inherently unreliable.

1.5 Average effects and agreement with parametric risk models

Each primary contrast uses two chains, with 2,000 burn-in and 1,000 retained draws per chain. For the exposed-group ratio-of-rates ERR, $\widehat{R}$ ranges from 1.000 to 1.025 and effective sample sizes from 309 to 1,382. For the EAR, effective sample sizes range from 276 to 1,617. The largest dispersion-parameter $\widehat{R}$ is 1.066 (stomach, high dose), indicating less complete mixing for that parameter. The negative-binomial dispersion posterior is approximately 157–190 for all-solid cancer and 21–30 for

Table 4: Marginal effects from adjusted parametric models by g-computation. Excess rates are cases per 10^4 person-years. Intervals are 95% confidence intervals.

Endpoint	Contrast	Model	Excess rate (95% CI)	Rate ratio (95% CI)
All solid	High	Poisson	32.6 (26.8, 38.3)	1.532 (1.435, 1.632)
	High	NB	32.7 (27.1, 38.5)	1.535 (1.440, 1.635)
	Medium	NB	7.35 (3.96, 10.70)	1.120 (1.064, 1.178)
	Low	NB	−0.11 (−2.60, 2.40)	0.998 (0.958, 1.040)
Stomach	High	NB	5.30 (2.57, 8.14)	1.319 (1.152, 1.495)
	Medium	NB	1.03 (−0.67, 2.80)	1.063 (0.961, 1.173)
	Low	NB	0.30 (−1.05, 1.62)	1.019 (0.940, 1.100)

stomach, with broad intervals (about 56–585 and 8.5–77 across the fits). Thus the all-solid count likelihood is close to Poisson conditional on the fitted baseline; stomach retains more extra-Poisson variation. These diagnostics describe the retained chains and do not establish interval calibration.

CountBCF and the adjusted negative-binomial generalised linear model agree in every marginal contrast (Table 5 and Figure 2). For high-dose all-solid cancer, CountBCF estimates an excess of 30.2 cases per 10^4 person-years and a rate ratio of 1.488, compared with 32.7 and 1.535 from the parametric model. The medium-dose contrast is positive under both methods, whereas the low-dose group is uninformative as an overall contrast. CountBCF lies nearer the null at medium and high dose, but this ordering does not hold in the low-dose comparisons. The exposed-group and cohort-averaged high-dose all-solid estimates are close: 30.37 and 30.22 excess cases per 10^4 person-years, respectively. Sections 1.7 and 1.14 use the exposed-group version to summarize risk and attributable burden among exposed survivors.

Table 5: Marginal effects from CountBCF and the adjusted negative-binomial GLM. Excess rates are cases per 10^4 person-years and ERR equals the rate ratio minus one. Intervals are 95% credible intervals for CountBCF and 95% confidence intervals for the GLM.

Endpoint	Contrast	Method	Excess rate (95% interval)	ERR (95% interval)	Rate ratio (95% interval)
All solid	High	CountBCF	30.2 (23.9, 35.9)	0.488 (0.381, 0.584)	1.488 (1.381, 1.584)
	High	NB GLM	32.7 (27.1, 38.5)	0.535 (0.440, 0.635)	1.535 (1.440, 1.635)
	Medium	CountBCF	6.95 (3.22, 10.67)	0.112 (0.051, 0.173)	1.112 (1.051, 1.173)
	Medium	NB GLM	7.35 (3.96, 10.70)	0.120 (0.064, 0.178)	1.120 (1.064, 1.178)
	Low	CountBCF	−0.32 (−2.90, 2.26)	−0.005 (−0.046, 0.037)	0.995 (0.954, 1.037)
	Low	NB GLM	−0.11 (−2.60, 2.40)	−0.002 (−0.042, 0.040)	0.998 (0.958, 1.040)
Stomach	High	CountBCF	4.28 (1.64, 7.11)	0.252 (0.095, 0.426)	1.252 (1.095, 1.426)
	Medium	CountBCF	0.22 (−1.58, 2.09)	0.014 (−0.087, 0.125)	1.014 (0.913, 1.125)
	Low	CountBCF	−0.59 (−2.03, 0.87)	−0.033 (−0.111, 0.051)	0.967 (0.889, 1.051)

For the 0.5–2 Gy group versus the < 5 mGy reference, the cohort-averaged ERR is 0.488 (0.381, 0.584) and the exposed-group ERR is 0.487 (0.386, 0.580). The person-year-weighted mean dose contrast is 0.9207 Gy. Dividing the latter ERR by this contrast gives 0.529 (0.419, 0.630) per Gy. This is a descriptive rescaling of a dose-category comparison, not a fitted continuous-dose slope.

The rescaled value lies near the published 0.47 per Gy at attained age 70 and exposure age 30 for 1958–1998 [Preston et al., 2007], but the averaging populations and covariate profiles differ. The published interval is a 90% confidence interval, (0.40, 0.54), whereas ours is a 95% credible interval. The extended 1958–2009 analyses report 0.50 (95% CI 0.42, 0.59) [Grant et al., 2017] and 0.510 (0.414, 0.612) under the reference-group analysis of French et al. [2017]; the different follow-up and

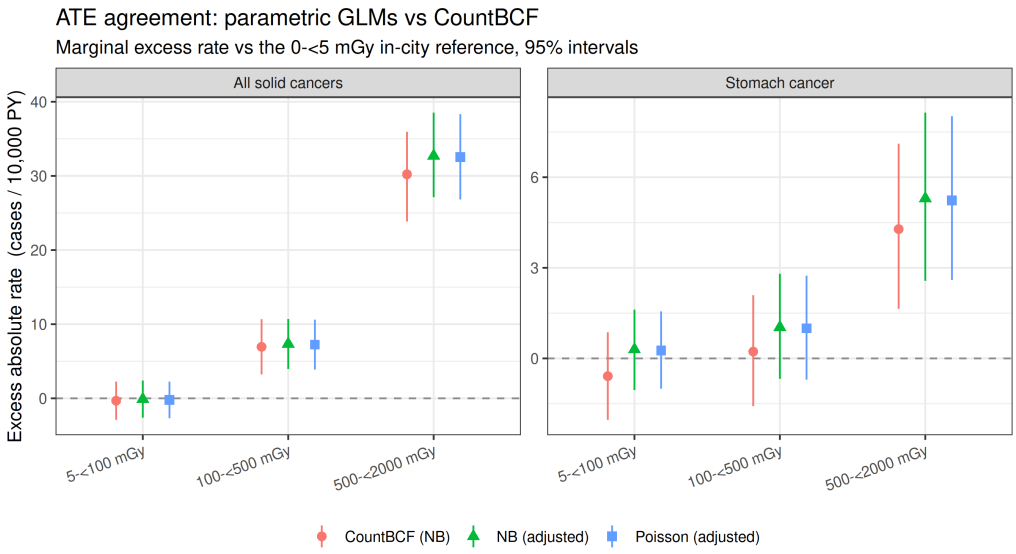


Figure 2: Convergence of marginal effects from CountBCF and the adjusted parametric model. Each estimate is a person-year-weighted excess rate with a 95% interval.

risk specifications also limit direct comparison. For medium-dose all-solid cancer and high-dose stomach cancer the analogous rescalings are 0.39 (0.15, 0.65) and 0.26 (0.10, 0.44) per Gy. The exposed-group ERRs −0.019, 0.092, and 0.487, at dose contrasts of 0.030, 0.234, and 0.921 Gy, recover the broad increasing risk pattern without imposing linearity. The low-dose estimate has too little precision to establish the shape near zero.

Table 6 gives the corresponding E-value sensitivity analysis [VanderWeele and Ding, 2017]. For the high-dose all-solid-cancer contrast, an unmeasured confounder would need associations of at least 2.34 with both dose-group membership and cancer incidence, on the rate-ratio scale, to explain the point estimate away; associations of at least 2.12 with both would be needed to move the interval limit nearest the null to the null. These values quantify sensitivity under the E-value assumptions; they do not establish exchangeability or rule out geographic confounding. The medium-dose all-solid result is less robust (E-values 1.41 and 1.23), and the high-dose stomach result is intermediate (1.81 and 1.42). For the low-dose contrasts, the interval E-value is one because the corresponding interval already contains the null. The E-values therefore reinforce the evidence for the high-dose all-solid association while leaving low-dose and site-specific inferences appropriately more tentative.

1.5.1 Fixed-age profile comparison

We also fitted the high-dose all-solid model with a longer sampling budget and evaluated it at exposure age 30 and attained age 70, separately by sex (Table 7). Within each sex, counterfactual rates were averaged across cities using the person-year shares of the exposed group before taking their ratio. The result remains a 0.5–2 Gy versus < 5 mGy group contrast. This is a separate fit from the primary run underlying the exposed-group ERR of 0.487; its two chain-specific exposed-group means are 0.468 and 0.469. Dividing each fixed-profile band ERR by the observed sex-specific mean dose contrast, 0.945 Gy for males and 0.906 Gy for females, provides a descriptive per-Gy rescaling rather than the continuous-dose coefficient estimated by the Preston model [Preston et al., 2007].

Two chains each used 5,000 burn-in iterations followed by 10,000 sampling iterations, retaining

Table 6: E-value sensitivity analysis for CountBCF. The rate ratios are exposed-group, person-year-weighted estimates (ATT), which differ slightly from the population-average rate ratios in Table 5. The point and interval E-values give the minimum strength of association, on the rate-ratio scale, that an unmeasured confounder would need with both treatment and outcome to explain the point estimate or move the interval nearest the null to the null.

Endpoint	Contrast	Exposed-group RR (95% CrI)	E-value, point	E-value, interval
All solid	Low	0.981 (0.940, 1.023)	1.16	1.00
	Medium	1.092 (1.035, 1.152)	1.41	1.23
	High	1.487 (1.386, 1.580)	2.34	2.12
Stomach	Low	0.956 (0.877, 1.040)	1.27	1.00
	Medium	1.002 (0.900, 1.112)	1.04	1.00
	High	1.252 (1.095, 1.426)	1.81	1.42

every tenth draw. Despite this budget, between-chain agreement at the fixed profiles was incomplete. For the city-pooled male and female band ERRs, rank-normalized split $\widehat{R}$ was 1.060 and 1.144, bulk effective sample size was 24 and 9, and tail effective sample size was 1,667 and 79, respectively. The Monte Carlo standard errors of their estimated means were 0.042 and 0.076. The tabulated means and quantiles of retained draws are therefore provisional MCMC summaries, rather than established posterior means and credible intervals. Further convergence work is needed. Their broad ranges and lower point estimates do not settle agreement with the published sex-specific risks, or establish a particular age-at-exposure pattern.

Table 7: Provisional MCMC summaries for all-solid cancer at exposure age 30 and attained age 70. CountBCF averages counterfactual rates across cities within sex and reports a dose-group ERR and its descriptive rescaling per Gy. CountBCF parentheses give empirical 2.5% and 97.5% quantiles of pooled retained draws, whose unresolved between-chain disagreement prevents their interpretation as reliable 95% credible intervals. Refitted and published Preston intervals are 90% confidence intervals. The Preston columns are continuous-dose ERR coefficients. Between-chain disagreement remains unresolved for the CountBCF profiles, so these summaries do not establish a stable fixed-profile comparison.

Sex	CountBCF band ERR	CountBCF ERR/Gy descriptive rescaling	Refitted Preston ERR/Gy	Published Preston ERR/Gy
Male	0.184 (−0.159, 0.623)	0.195 (−0.168, 0.660)	0.336 (0.245, 0.426)	0.35 (0.28, 0.43)
Female	0.315 (−0.095, 0.845)	0.348 (−0.105, 0.933)	0.554 (0.438, 0.670)	0.58 (0.43, 0.69)

1.6 Heterogeneity of the radiation-effect surface

CountBCF estimates effect modification without specifying interaction functions. A subgroup is called separated when the 95% posterior interval for its ratio-of-rates ERR minus the cohort-averaged ERR excludes zero. Unless stated otherwise, both quantities average over all cells in the corresponding two-dose-group comparison. The cell-average ERR is displayed separately and does not determine this indicator. The EAR describes absolute burden. Subgroup differences are therefore assessed on a fixed summary scale and averaging population, using their joint posterior draws.

The six endpoint-by-dose-group fits generate 294 subgroup comparisons across nine overlapping subgroup partitions per fit. Their counts are reported in Table 8: 34 intervals exclude zero on the ratio-of-rates scale and 38 on the cell-average scale, with 27 of the former arising in the high-dose all-solid analysis. These posterior intervals are not multiplicity adjusted and the comparisons are dependent. We therefore assess them alongside the regularisation sweep in Section 1.8 and data perturbations in Section 1.9. A conventional parametric check on the same data supplies an additional model comparison, not independent replication.

Table 8: Multiplicity accounting for the subgroup separation test. Every test in the six primary fits is counted. A test indicates separation when the 95% credible interval for the subgroup effect minus the cohort average effect excludes zero.

Fit	Separation tests	Separated, ratio of rates	Separated, per-cell
All solid, 500–< 2000 mGy	49	27	26
All solid, 100–< 500 mGy	49	2	5
All solid, 5–< 100 mGy	49	5	7
Stomach, 500–< 2000 mGy	49	0	0
Stomach, 100–< 500 mGy	49	0	0
Stomach, 5–< 100 mGy	49	0	0
Total	294	34	38

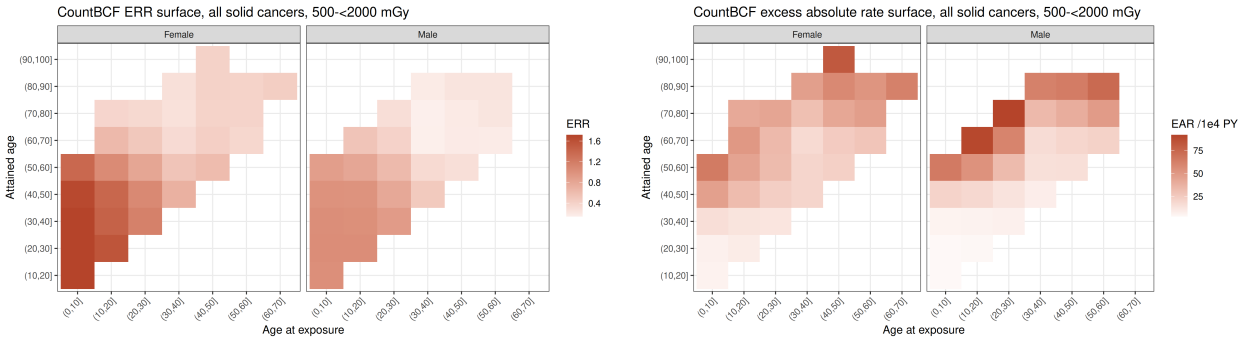
1.6.1 How much heterogeneity is present at all

Before examining named subgroups it is useful to ask how much variation the posterior surface contains. Table 9 reports the posterior person-year-weighted standard deviation of the per-cell ERR across exposed cells, together with the share of exposed cells whose effect is positive and the share whose 95% interval excludes zero.

Table 9: Cell-level heterogeneity of the CountBCF effect surface. The first column is the posterior person-year-weighted standard deviation of the per-cell ERR across exposed cells.

Endpoint	Contrast	SD of per-cell ERR (95% CrI)	Cells with ERR > 0	Cells with CrI excluding 0
All solid	High	0.584 (0.416, 0.773)	98.6%	63.0%
	Medium	0.253 (0.198, 0.324)	82.8%	2.0%
	Low	0.233 (0.183, 0.299)	58.1%	2.4%
Stomach	High	0.004 (0.000, 0.030)	100%	100%
	Medium	0.146 (0.076, 0.240)	73.5%	0%
	Low	0.186 (0.135, 0.257)	54.3%	0%

Across exposed high-dose all-solid cells, posterior mean ERRs have median 0.60 and a tenth-to-ninetieth percentile range of 0.16 to 1.57. These quantiles describe variation across cells, rather than posterior uncertainty in one effect. Nearly two thirds of individual cell intervals exclude zero. At medium and low dose the surface is much flatter and almost no individual cell separates, which is a statement about power rather than about the absence of modification. For stomach at high dose the moderating forest returns an essentially constant effect, an ERR of 0.252 in every subgroup, with posterior mean weighted surface standard deviation 0.004; that case is discussed in Section 1.6.4. Figures 3a and 3b display the fitted relative and absolute surfaces over the covariate space.



(a) Relative effect (ERR).

(b) Absolute effect (EAR).

Figure 3: Estimated CountBCF effect surfaces for high-dose all-solid cancer over the observed covariate space. The relative and absolute surfaces move in opposite directions with attained age.

1.6.2 All solid cancers at 500–< 2000 mGy

Table 10 gives the complete set of high-dose subgroup effects with their separation tests. Five conclusions follow.

First, age at exposure is the dominant modifier of the relative effect. The ratio-of-rates ERR declines more than fourfold, from 1.26 (0.85, 1.71) among those exposed at ages 0–9 to 0.29 (0.14, 0.44) among those exposed at age 40 or later, and both tails separate credibly from the average. This is the textbook LSS result, recovered with no interaction term specified anywhere.

Second, attained age modifies the relative and absolute effects in opposite directions. The ratio-of-rates ERR falls from 1.40 (0.97, 1.84) below age 40 to 0.30 (0.11, 0.51) at age 80 or older, while the EAR rises from 8.7 to 57.3 excess cases per 10^4 person-years, and every attained-age group separates. The two are projections of a single surface, since the EAR is approximately the baseline rate multiplied by the ERR and the baseline rate climbs faster with attained age than the relative effect fades.

Third, time since exposure moves the absolute effect but not the relative one. No clear variation of the ERR is detected across the groups spanning 20 to more than 40 years of follow-up, while the EAR nearly quadruples from 15.2 to 57.3 per 10^4 person-years. These group summaries provide little evidence about changes in the relative excess; the rising absolute burden reflects the larger baseline rates.

Fourth, the cohort-averaged high-dose ERR is 0.58 (0.45, 0.72) in women and 0.38 (0.25, 0.51) in men, a female-to-male ratio of 1.54. Over exposed cells the corresponding values are 0.603 and 0.369, a ratio of 1.63, close to the published 1.6. Using unrounded posterior means, the female-minus-male difference is 0.204 over all comparison cells and 0.233 over exposed cells; subtracting rounded table entries gives 0.205 and 0.234, respectively. The sensitivity analysis uses a separate shorter single-chain fit, yielding 0.256; the different averaging population and fit explain why these values differ across tables. Its magnitude is sensitive to the prior, and its interval includes zero when breast and thyroid are removed (Sections 1.8 and 1.9).

Fifth, the cohort-averaged city ERRs are 0.49 (0.37, 0.60) in Hiroshima and 0.48 (0.31, 0.67) in Nagasaki. The exposed-group Nagasaki-minus-Hiroshima difference from the sensitivity fit is −0.052 (−0.247, 0.154). This interval is compatible with substantial differences in either direction, including the low-dose contrast. It neither establishes equality at high dose nor constrains explanations of the low-dose city association.

These patterns match the central qualitative findings of LSS incidence modelling: relative risk

Table 10: High-dose all-solid-cancer conditional effects (500–< 2000 mGy against the 0–< 5 mGy reference). The first three numeric columns are on the ratio-of-rates scale, which is the one that determines every claim: a subgroup is marked as separated when the 95% posterior interval for its ratio-of-rates ERR minus the cohort-averaged ERR of 0.488 excludes zero. The final column is the person-year-weighted mean of the per-cell ERR, an average of ratios, given for cross-reference only; it runs systematically larger and is taken against a different reference, the per-cell ATE of 0.87. The Cases column counts cases in the subgroup across both dose groups.

Dimension	Subgroup	Cases	Group ERR (95% CrI)	CATE − ATE, ratio of rates (95% CrI)	Sep.?	Per-cell ERR
Sex	Male	3,089	0.38 (0.25, 0.51)	−0.11 (−0.21, −0.02)	Yes	0.74
	Female	3,662	0.58 (0.45, 0.72)	+0.09 (0.01, 0.18)	Yes	0.94
Age at exposure	0–9	532	1.26 (0.85, 1.71)	+0.77 (0.39, 1.18)	Yes	1.45
	10–19	1,303	0.80 (0.60, 1.02)	+0.31 (0.14, 0.50)	Yes	1.08
	20–29	1,159	0.51 (0.33, 0.71)	+0.02 (−0.15, 0.19)	No	0.72
	30–39	1,499	0.27 (0.10, 0.45)	−0.22 (−0.37, −0.06)	Yes	0.37
	40+	2,258	0.29 (0.14, 0.44)	−0.20 (−0.32, −0.08)	Yes	0.32
Attained age	< 40	167	1.40 (0.97, 1.84)	+0.91 (0.52, 1.33)	Yes	1.41
	40–59	1,677	0.91 (0.71, 1.12)	+0.42 (0.27, 0.59)	Yes	1.00
	60–79	3,818	0.33 (0.22, 0.44)	−0.16 (−0.23, −0.09)	Yes	0.36
	80+	1,089	0.30 (0.11, 0.50)	−0.19 (−0.35, −0.02)	Yes	0.32
City	Hiroshima	4,369	0.49 (0.37, 0.60)	+0.004 (−0.07, 0.08)	No	0.77
	Nagasaki	2,382	0.48 (0.31, 0.67)	−0.007 (−0.14, 0.13)	No	1.02
Time since exposure	< 20 y	939	0.45 (0.29, 0.62)	−0.04 (−0.16, 0.10)	No	0.93
	20–29 y	1,430	0.44 (0.32, 0.56)	−0.05 (−0.13, 0.03)	No	0.90
	30–39 y	1,670	0.49 (0.38, 0.60)	+0.002 (−0.04, 0.04)	No	0.88
	40+ y	2,712	0.53 (0.39, 0.66)	+0.04 (−0.05, 0.13)	No	0.75

Excess absolute rates per 10^4 person-years for the same subgroups, in the order listed, are 28.9 and 31.0 by sex; 24.6, 34.3, 31.8, 24.4, and 35.5 by age at exposure; 8.7, 34.3, 37.6, and 57.3 by attained age; 33.0 and 26.2 by city; and 15.2, 20.8, 32.1, and 57.3 by time since exposure.

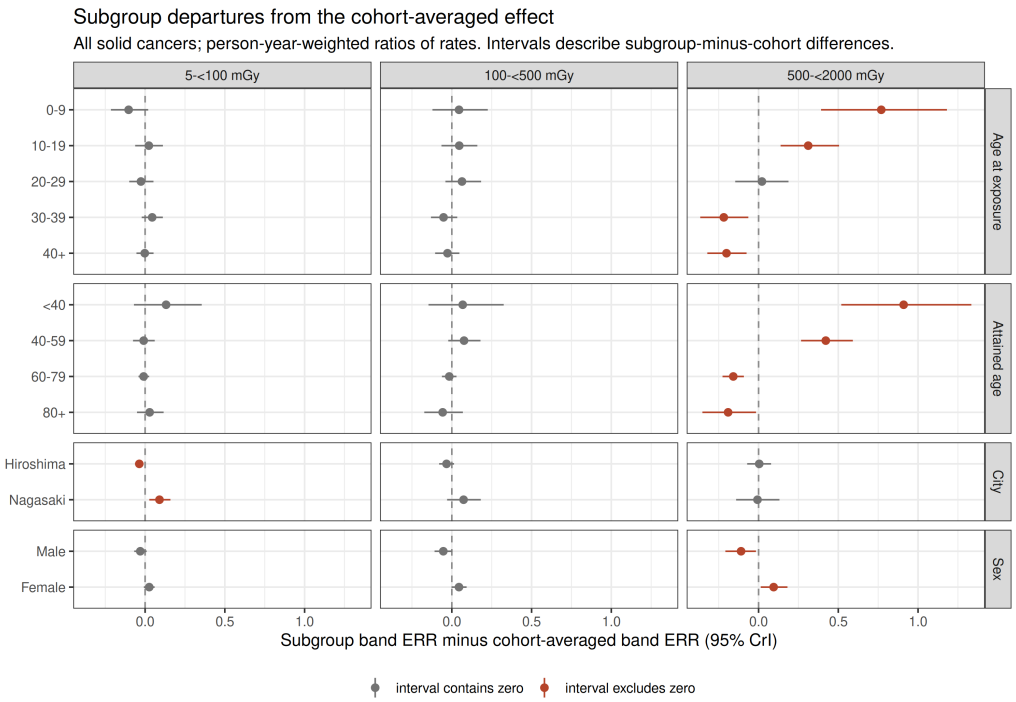


Figure 4: Subgroup band ERR differences in all three all-solid-cancer dose contrasts. Each point is a posterior subgroup ratio-of-rates ERR minus its cohort-averaged band ERR, with a pointwise 95% credible interval. The intervals describe exploratory subgroup comparisons without a multiplicity correction.

is greater following younger-age exposure, while the absolute rate burden grows at older attained ages [Preston et al., 2007, Grant et al., 2017].

1.6.3 All solid cancers at 100–< 500 mGy

On the ratio-of-rates scale no one-dimensional subgroup separates at 100–< 500 mGy. The sex contrast comes closest, with a male ratio-of-rates ERR of 0.058 (−0.021, 0.143) and a difference from the average of −0.054 (−0.108, 0.003), against a female 0.156 (0.079, 0.240) and +0.045 (−0.002, 0.093); both intervals just cover zero. On the per-cell summary scale the same contrast is 0.061 against 0.222 and does separate, at −0.101 (−0.200, −0.002) and +0.059 (0.001, 0.118), which is a difference of summary scale rather than of evidence: the medium-dose sex effect is borderline either way and is not featured. No one-dimensional age-at-exposure or attained-age subgroup separates at this dose. Two two-dimensional comparisons do separate on the ratio-of-rates scale: women at attained ages 40–59, and Hiroshima men. These exploratory intersections account for the two medium-dose entries in Table 8.

The pattern across dose groups is nevertheless informative, and it points at something the canonical model already contains. The female-to-male ratio of ratio-of-rates ERRs is about 2.7 at 100–< 500 mGy and 1.54 at 500–< 2000 mGy, so the sex gap is widest where the male response is weakest in absolute terms. That is what the published curvature of the male dose response implies. Grant et al. [2017] report upward curvature in the male dose response, including an apparent flattening of the response between roughly 0.2 and 0.7 Gy; French et al. [2017] report strong evidence of a curvilinear dose response among males, at $P = 0.008$ in their reference-group analysis; and Preston et al. [2007] already reported a borderline sex-averaged test for quadratic departure from linearity over 0–2 Gy, at $P = 0.09$, with an estimated curvature of 0.3 (90% CI 0.01, 0.90). A male response that is flat across the medium-dose group and then rises steeply depresses the male ERR at 100–< 500 mGy relative to a linear expectation and restores it at 500–< 2000 mGy, which is qualitatively compatible with the dose-group pattern above. These broad categories and uncertain sex contrasts do not establish curvature in the CountBCF dose response.

1.6.4 Stomach cancer as a counterpoint

Stomach cancer provides a lower-information comparison. Its high-dose average effect is positive, but the moderating forest is effectively constant: the cohort-averaged, exposed-group, and cell-specific ERRs are all approximately 0.2519. The posterior surface standard deviation is 0.004. This is absence of detected modification under the fitted prior, rather than evidence that the true effect is homogeneous. There are 269 exposed stomach cases. Published tests give $P = 0.4$ for age at exposure and $P = 0.05$ for attained age, which likewise leave considerable uncertainty. The stomach regularisation sweep in Table 13 examines the sensitivity of this shrinkage; an extreme shared-prior setting gives an unstable heterogeneous surface.

1.7 Comparison with the canonical LSS surface

The average-effect comparison is only a partial validation. We therefore evaluate the CountBCF surface against the refitted risk model of Section 1.4.1 at the same person-year cells. Two comparisons are possible. The ratio-of-rates comparison weights each surface by the same CountBCF-estimated reference rate. It holds the baseline weighting fixed while comparing the two relative-effect surfaces. The per-cell comparison is the plain person-year-weighted mean of the cell ERRs; it is systematically larger for both surfaces and addresses a different averaging target. We report the ratio-of-rates comparison. Unlike the separation tests of Section 1.6, which average the conditional effect over

the whole cohort, this comparison is evaluated only at the exposed person-year cells, because the parametric ERR is proportional to dose and collapses to zero at the reference cells whereas the CountBCF contrast does not; the two surfaces can therefore only be compared where the exposure actually occurred. A given subgroup's CountBCF ERR here can accordingly differ by a few percent from its cohort-averaged value in Table 10.

1.7.1 All solid cancers

On the ratio-of-rates scale the CountBCF-minus-Preston intervals contain zero in 26 of 28 high-dose, 25 of 28 medium-dose, and 22 of 28 low-dose subgroups (Table 11 and Figure 5). Their person-year-weighted mean EAR is numerically close at high dose, 30.37 per 10^4 person-years for each model. Across the 5,681 exposed cells the two surfaces correlate at 0.57 on the Pearson scale and 0.80 on the Spearman scale. CountBCF was supplied neither a linear dose term nor any interaction involving age, sex, or city. These results support recovery of broad risk patterns without specifying those interactions. The selected, overlapping subgroups do not establish equivalence across the full covariate space, and uncertainty in the fitted Preston parameters is not propagated.

Table 11: Ratio-of-rates comparison of the CountBCF surface with the refitted canonical risk model, evaluated at the same person-year cells. Every subgroup whose 95% credible interval for the difference excludes zero is listed, for all three dose groups, together with the overall row and the two sexes at high dose for context. Rows marked † are the departures.

Contrast	Subgroup	Refitted model ERR	CountBCF ERR (95% CrI)	Difference (95% CrI)
High	Overall	0.553	0.487 (0.386, 0.580)	−0.066 (−0.168, 0.026)
	Attained age < 40 †	2.759	1.330 (0.931, 1.758)	−1.429 (−1.863, −0.965)
	Time since exposure < 20 y †	0.659	0.471 (0.301, 0.640)	−0.188 (−0.356, −0.015)
	Age at exposure 0–9	1.522	1.184 (0.794, 1.607)	−0.338 (−0.735, 0.090)
	Female	0.707	0.603 (0.465, 0.741)	−0.104 (−0.243, 0.035)
	Male	0.396	0.369 (0.240, 0.500)	−0.027 (−0.157, 0.104)
Medium	Overall	0.138	0.092 (0.035, 0.152)	−0.046 (−0.103, 0.013)
	Age at exposure 0–9 †	0.368	0.114 (−0.053, 0.294)	−0.253 (−0.421, −0.073)
	Attained age < 40 †	0.736	0.142 (−0.072, 0.389)	−0.593 (−0.813, −0.340)
	Female exposed at 0–9 †	0.508	0.193 (−0.016, 0.440)	−0.315 (−0.531, −0.070)
Low	Overall	0.019	−0.019 (−0.060, 0.023)	−0.038 (−0.078, 0.005)
	Hiroshima †	0.019	−0.043 (−0.089, 0.003)	−0.063 (−0.108, −0.016)
	Nagasaki	0.017	0.088 (0.006, 0.171)	+0.072 (−0.011, 0.154)
	Age at exposure 0–9 †	0.044	−0.141 (−0.256, −0.015)	−0.185 (−0.299, −0.060)
	Time since exposure 40+ y †	0.018	−0.059 (−0.115, −0.001)	−0.077 (−0.134, −0.019)
	Male exposed at 0–9 †	0.031	−0.143 (−0.279, 0.010)	−0.174 (−0.309, −0.022)
	Female exposed at 0–9 †	0.059	−0.136 (−0.274, 0.015)	−0.195 (−0.333, −0.045)
	Male exposed at 20–29 †	0.013	−0.110 (−0.222, 0.011)	−0.123 (−0.235, −0.002)

The low-dose block deserves a separate remark, because it is the one that bears on Section 1.10. Six of the 28 low-dose subgroups depart from the canonical surface on the ratio-of-rates scale, and every one of them departs downward: CountBCF puts the low-dose-group ERR below what a linear dose response extrapolated from the whole 0–2 Gy range predicts. Hiroshima is among the six, at −0.063 (−0.108, −0.016). Nagasaki is not, at +0.072 (−0.011, 0.154). On the scale that this study uses for every other claim, therefore, Nagasaki's low-dose-group ERR is not distinguishable from what the canonical model already predicts, and Hiroshima's is. Whatever the low-dose city contrast of Section 1.10 is, it is at least as much a statement about Hiroshima sitting below the canonical prediction as about Nagasaki sitting above it.

The disagreements are localized rather than scattered. They occur at young attained age and

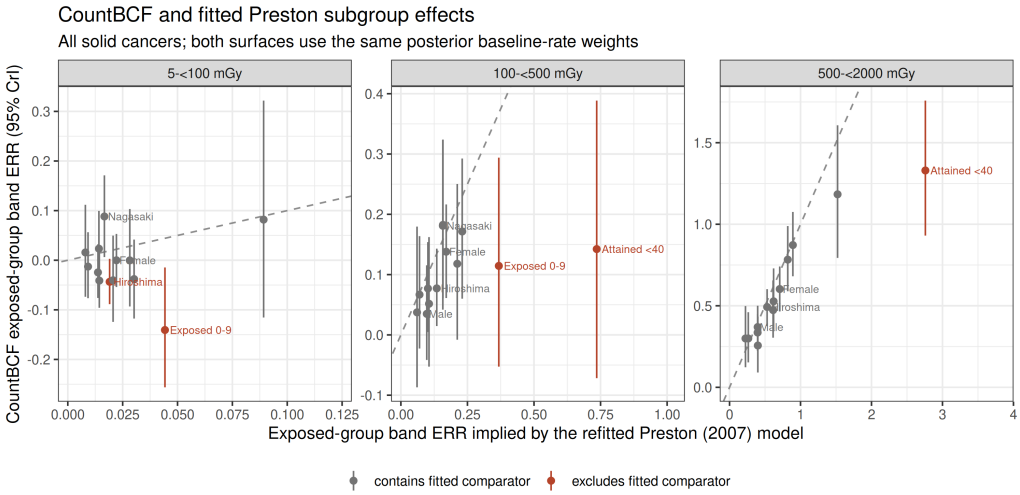


Figure 5: Exposed-group ratio-of-rates ERR summaries compared with the refitted Preston model, using the same CountBCF posterior baseline-rate weights for both summaries. Bars show CountBCF 95% credible intervals; uncertainty in the fitted Preston parameters is not propagated. Most examined subgroup intervals contain the fitted comparator, with larger departures in some young-age groups. Labels identify selected groups and distinguish age at exposure from attained age.

short time since exposure, where the parametric attained-age power law extrapolates most strongly. At attained age below 40 the refitted model predicts a high-dose ERR of 2.76 at 0.92 Gy, while CountBCF estimates 1.33 (0.93, 1.76) and excludes the parametric value. The same pattern appears one dose group down, at 0.74 against 0.14, and for stomach, at 2.89 against 0.25. A power law in attained age is forced to be monotone and unbounded as attained age tends to zero; the data in that corner are sparse, with only 51 exposed cases below attained age 40 in the high-dose group, and CountBCF's answer is a flattening rather than a divergence. The published analysis itself notes this region as unsettled, noting that a more complex pattern provides a somewhat better fit for all solid cancers and that future analyses should give more attention to alternative descriptions of age-at-exposure effects on both the ERR and the EAR [Preston et al., 2007]. The competing explanation has to be stated plainly: because the moderating-forest prior shrinks toward a constant, a flattening in a thin corner is also what excessive shrinkage would produce. Section 1.8 is the decisive test of that alternative.

1.7.2 Stomach cancer

For stomach at high dose, 21 of the 28 subgroups depart on the ratio-of-rates scale, and 23 of 28 on the per-cell scale, because the parametric model carries effect modification, with an ERR proportional to attained age to the power -1.5, a decline of 13% per decade of age at exposure, and a female-to-male ratio of 2.3, while CountBCF returns a flat ERR of 0.252. Published stomach tests give $P = 0.4$ for age at exposure and $P = 0.05$ for attained age. CountBCF detects no stomach subgroup separation under the default prior. The difference from the parametric surface reflects both functional-form assumptions and shrinkage in a dataset with limited site-specific information; it cannot identify which effect surface is closer to the truth. The stomach-specific sweep in Section 1.8, Table 13, detects no subgroup separation under all but one setting. The exception is an unstable fit with a large shared leaf-concentration prior.

1.8 Robustness to regularisation of the moderating forest

Because CountBCF deliberately tightens the prior on the treatment-effect surface in order to suppress spurious heterogeneity, any featured subgroup effect must be shown to survive both loosening and tightening that prior. Table 12 reports twelve refits of the featured contrast, all solid cancers at 500–< 2000 mGy, varying the number of moderating trees, the tree-depth penalty on the moderating forest, and the overall leaf concentration. With person-years as the offset, the data-dependent default formula is negative and the implementation falls back to $a_0 = 1$. The shared-prior sweep therefore uses $a_0 = 0.5$, 1, and 2, halving and doubling that fallback. This differs from the no-offset simulations, where the data-dependent formula is used. Figure 6 displays the same sweep.

Table 12: Sensitivity of the featured contrast (all solid cancers, 500–< 2000 mGy) to regularisation of the moderating forest. All entries are on the ratio-of-rates scale; the last two columns are posterior differences with 95% credible intervals. m_τ and the depth prior act on the moderating forest alone, whereas the leaf concentration a_0 is shared by both forests, so the a_0 rows are not a moderating-forest-only sweep. The twelve rows are ten distinct fits: the default specification is $m_\tau = 50$ with depth prior $(0.25, 3)$ and $a_0 \times 1$, so it is the common row of all three sweeps and is reported three times (‡).

Specification	Ratio-of-rates ERR (95% CrI)	Group ERR, exposed < 20 minus 40+	Group ERR, female minus male
$m_\tau = 10$ (tightest)	0.434 (0.345, 0.521)	0.421 (0.208, 0.612)	0.156 (0.003, 0.307)
$m_\tau = 25$	0.486 (0.389, 0.582)	0.529 (0.298, 0.787)	0.225 (0.051, 0.407)
$m_\tau = 50$ (default) ‡	0.495 (0.401, 0.594)	0.527 (0.281, 0.778)	0.256 (0.074, 0.431)
$m_\tau = 100$	0.508 (0.416, 0.608)	0.597 (0.344, 0.864)	0.223 (0.038, 0.414)
$m_\tau = 200$ (loosest)	0.505 (0.405, 0.605)	0.604 (0.356, 0.859)	0.269 (0.084, 0.456)
Depth prior (0.05, 4), tight	0.494 (0.400, 0.599)	0.563 (0.313, 0.813)	0.208 (0.029, 0.380)
Depth prior (0.25, 3), default ‡	0.495 (0.401, 0.594)	0.527 (0.281, 0.778)	0.256 (0.074, 0.431)
Depth prior (0.5, 2), loose	0.496 (0.398, 0.598)	0.574 (0.325, 0.824)	0.240 (0.054, 0.438)
Depth prior (0.95, 2), prognostic-like	0.489 (0.397, 0.585)	0.583 (0.338, 0.839)	0.260 (0.086, 0.445)
$a_0 \times 0.5$ (both forests)	0.465 (0.359, 0.576)	0.272 (0.125, 0.422)	0.029 (0.003, 0.057)
$a_0 \times 1$ (default) ‡	0.495 (0.401, 0.594)	0.527 (0.281, 0.778)	0.256 (0.074, 0.431)
$a_0 \times 2$ (both forests)	0.463 (0.367, 0.558)	0.579 (0.322, 0.831)	0.254 (0.061, 0.436)

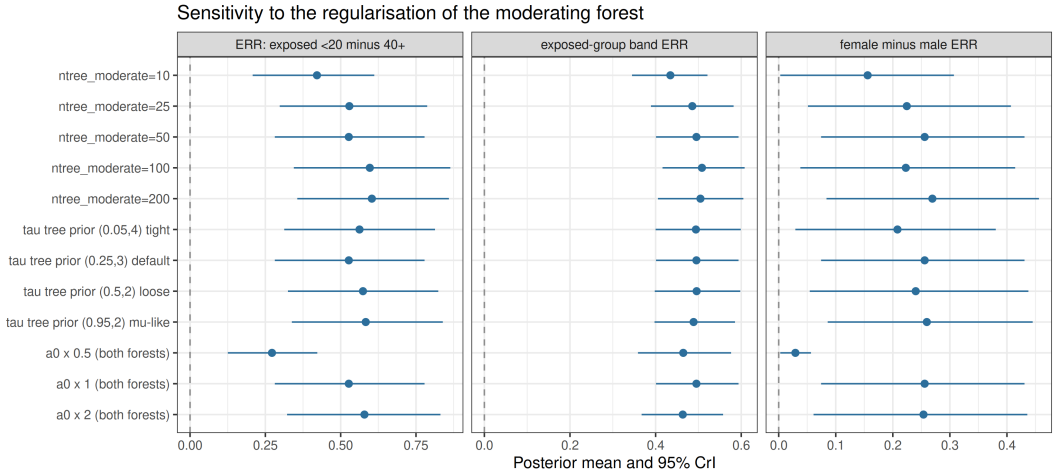


Figure 6: Sensitivity of the featured contrast to the moderating-forest prior. All three panels are on the ratio-of-rates scale, matching Table 12.

Across the twelve settings, the exposed-group ERR ranges from 0.434 to 0.508. The age-at-exposure contrast (exposed before age 20 versus at age 40 or later) retains a credible interval

excluding zero in every setting, with posterior means from about 0.27 to 0.60. The female-minus-male contrast also retains interval exclusion throughout, but its magnitude falls from 0.256 (0.074, 0.431) at the default to 0.029 (0.003, 0.057) with $a_0 = 0.5$. Its sign is stable within this sweep; its size is strongly prior sensitive. Section 1.9 shows that its interval also includes zero under several cancer-site exclusions.

The high-dose city-difference interval includes zero in eleven settings. At $a_0 = 0.5$ it is 0.029 (0.004, 0.054), with a similarly small and narrow sex contrast. These differences show that changing the shared prior affects both the shape and uncertainty of the fitted surface. They do not establish that the cities have equal effects under the remaining settings.

The same twelve-setting grid was then applied to the stomach high-dose contrast to test the mirror-image concern: could a modifier hidden by the default prior appear when the regularisation is changed? Table 13 reports the result. At the default, the posterior person-year-weighted standard deviation of the stomach ERR surface is 0.0042, none of the 49 subgroup tests separates on either the ratio-of-rates or per-cell summary scale, and the largest ratio-of-rates subgroup deviation from the average is 0.0002. The same zero separation result holds in the other eleven settings except for the extreme $a_0 \times 2$ setting, where the surface becomes unstable: its weighted standard deviation of cell ERRs is 1.14 and 9 of 49 ratio-of-rates tests (15 of 49 per-cell tests) separate. The overall effect also shifts to 0.184 (0.036, 0.356) there. The ordinary regularisation changes do not reveal stomach modification, whereas doubling the shared concentration gives a much more variable surface. This sensitivity and the limited number of exposed cases prevent a strong conclusion about the presence or absence of site-specific modification.

Table 13: Sensitivity of the stomach high-dose contrast to the regularisation grid used for the all-solid analysis. The ratio-of-rates ERR is the person-year-weighted exposed-group contrast with its 95% credible interval. The next two columns give the posterior person-year-weighted standard deviation of the per-cell ERR surface and the largest absolute ratio-of-rates subgroup deviation from its cohort average, respectively. The final columns count subgroup tests whose 95% interval excludes zero, out of 49 tests, on the ratio-of-rates and per-cell summary scales respectively. The a_0 rows act on both forests; the default is repeated across the three sweeps (‡).

Specification	Ratio-of-rates ERR (95% CrI)	SD(ERR)	Max $\lvert d \rvert$	Group indicators / 49	Per-cell indicators / 49
$m_\tau = 10$ (tightest)	0.226 (0.078, 0.390)	0.0007	0.00004	0	0
$m_\tau = 25$	0.242 (0.084, 0.417)	0.0026	0.00014	0	0
$m_\tau = 50$ (default) ‡	0.253 (0.078, 0.429)	0.0042	0.00019	0	0
$m_\tau = 100$	0.255 (0.082, 0.427)	0.0075	0.00038	0	0
$m_\tau = 200$ (loosest)	0.260 (0.089, 0.452)	0.0124	0.00063	0	0
Depth prior $(0.05, 4)$, tight	0.252 (0.080, 0.422)	0.0008	0.00010	0	0
Depth prior $(0.25, 3)$, default ‡	0.253 (0.078, 0.429)	0.0042	0.00019	0	0
Depth prior $(0.5, 2)$, loose	0.253 (0.088, 0.416)	0.0090	0.00061	0	0
Depth prior $(0.95, 2)$, extreme	0.248 (0.087, 0.419)	0.0414	0.00655	0	0
$a_0 \times 0.5$ (both forests)	0.234 (0.085, 0.397)	0.0754	0.053	0	0
$a_0 \times 1$ (default) ‡	0.253 (0.078, 0.429)	0.0042	0.00019	0	0
$a_0 \times 2$ (both forests)	0.184 (0.036, 0.356)	1.136	1.668	9	15

1.9 Robustness to the data construction

Table 14 reports fourteen refits of the featured contrast, each changing one element of the data construction, with Figure 7 showing the same fits.

Adding the 1,285 not-in-city cells to the reference group with an indicator in the prognostic forest moves the ratio-of-rates ERR from 0.495 to 0.501, a change of 1.2%. This is small relative to

Table 14: Sensitivity of the featured contrast to the data construction. Each row changes one element relative to the baseline. All ERR entries are on the ratio-of-rates scale fixed in Section 1.6; the last two columns are posterior differences with 95% credible intervals.

Perturbation	Cells	Cases	Ratio-of-rates ERR (95% CrI)	EAR/10^4 PY	Group ERR, exposed $<$ 20 minus 40+	Group ERR, female minus male
Baseline (all solid, colon-dose groups, NB)	8,249	6,751	0.495 (0.401, 0.594)	30.7	0.527 (0.281, 0.778)	0.256 (0.074, 0.431)
Poisson likelihood	8,249	6,751	0.498 (0.402, 0.598)	30.8	0.563 (0.311, 0.819)	0.240 (0.050, 0.413)
Not in city added to reference group	9,534	10,745	0.501 (0.408, 0.603)	30.9	0.552 (0.299, 0.799)	0.233 (0.059, 0.417)
Reference widened to 0–$<$ 100 mGy	14,222	11,157	0.503 (0.406, 0.606)	31.0	0.567 (0.334, 0.807)	0.237 (0.075, 0.411)
High group extended to include $\geq$ 2 Gy	10,714	6,936	0.588 (0.494, 0.691)	35.9	0.713 (0.456, 0.986)	0.289 (0.096, 0.480)
Excluding breast and thyroid	8,249	6,137	0.405 (0.310, 0.510)	23.1	0.287 (0.049, 0.534)	0.066 (−0.112, 0.261)
Excluding non-melanoma skin	8,249	6,605	0.479 (0.384, 0.576)	29.0	0.491 (0.253, 0.752)	0.210 (0.035, 0.398)
Excluding stomach	8,249	4,975	0.575 (0.458, 0.697)	25.9	0.532 (0.249, 0.825)	0.206 (−0.030, 0.412)
Non-sex-specific solid only	8,249	5,502	0.525 (0.416, 0.642)	26.5	0.420 (0.161, 0.689)	0.142 (−0.069, 0.336)
Stomach, colon-dose groups	8,249	1,776	0.253 (0.078, 0.429)	4.41	−0.0001 (−0.003, 0.002)	0.00002 (−0.0004, 0.0005)
Stomach, stomach-dose groups	8,075	1,774	0.256 (0.092, 0.443)	4.45	−0.0001 (−0.003, 0.002)	0.00001 (−0.0004, 0.0005)
Lung, lung-dose groups	7,636	683	0.666 (0.356, 1.001)	4.21	−0.047 (−0.472, 0.272)	0.480 (0.002, 1.086)
Colon, colon-dose groups	8,249	584	0.297 (0.019, 0.586)	1.64	0.052 (−0.211, 0.335)	−0.095 (−0.473, 0.242)
Liver, liver-dose groups	7,917	592	0.291 (0.024, 0.609)	1.66	0.073 (−0.185, 0.396)	−0.121 (−0.455, 0.193)

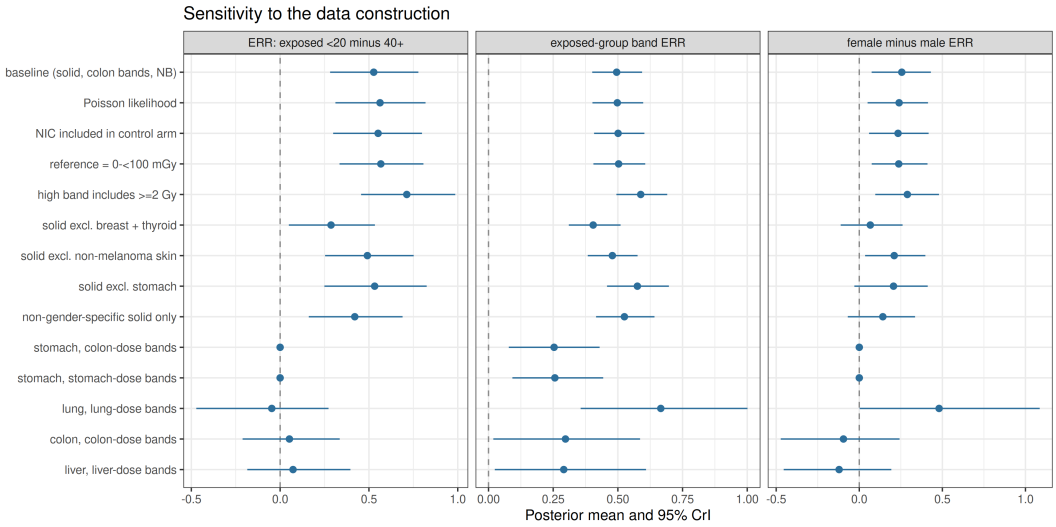


Figure 7: Sensitivity of the featured contrast to the data construction. All three panels are on the ratio-of-rates scale, matching Table 14. Across the nine all-solid variants, the average effect and the age-at-exposure contrast retain positive intervals. The separate site-specific fits need not show the same age contrast. The largest all-solid movements are the removal of breast and thyroid, which lowers both, and the extension of the high-dose group beyond 2 Gy, which raises both. The sex difference behaves differently, losing its separation from zero in three of the four all-solid composition variants.

the posterior interval width and is consistent with the limited sensitivity to NIC inclusion reported by Preston et al. [2007].

Changing the reference group or dose cap also changes the binary contrast. Widening the reference group to 0–< 100 mGy gives an ERR of 0.503 against 0.495. Restoring survivors at or above 2 Gy raises it to 0.588 as the exposed group's mean dose increases. The descriptive per-Gy rescalings are similar, without establishing a linear dose response. Poisson and negative-binomial likelihoods give 0.498 and 0.495 for the original contrast, consistent with the near-Poisson dispersion posterior of Section 1.5.

Cancer-site exclusions show the sensitivity of the sex contrast left open in Section 1.8. Removing breast and thyroid drops the ratio-of-rates ERR from 0.495 to 0.405, a fall of 18%, and the ratio-of-rates female-minus-male difference collapses from 0.256 (0.074, 0.431) to 0.066 (−0.112, 0.261), its interval now covering zero. Removing non-melanoma skin, by contrast, changes almost nothing, giving 0.479 with an age-at-exposure gradient of 0.491 against the baseline 0.527, so the observed age gradient does not depend on including non-melanoma skin although that site has the steepest age-at-exposure effect in the published heterogeneity tests. Much of the estimated sex contrast depends on the inclusion of breast and thyroid, consistent with the published reduction in the sex difference when the analysis is restricted to non-sex-specific cancers on the EAR scale. Our non-sex-specific fit leaves a difference of 0.142 (−0.069, 0.336), attenuated by about 45% and no longer separating from zero, although the point estimate stays positive in the direction implied by the published ERR sex ratio of 1.8 for non-sex-specific cancers.

This is what disqualifies the sex difference from the featured results. Its credible interval excludes zero under every prior in Table 12, but it does not survive the data: three of the four all-solid composition variants extinguish it, the exception being the removal of non-melanoma skin. The contrast with the age-at-exposure gradient is the point. That gradient survives every data perturbation for all-solid cancer, with a credible interval excluding zero in all nine variants, including the removal of breast and thyroid, and a magnitude between 0.29 and 0.71. A subgroup effect worth featuring has to hold under both kinds of perturbation, and only one of the two does.

The site-specific results are internally consistent and match the canon where it is sharp. Lung gives a ratio-of-rates ERR of 0.666 with a credible female-minus-male ERR difference of 0.480 (0.002, 1.086), against published findings of the largest sex effects for lung with female ERRs three to five times the male values. Colon at 0.297 and liver at 0.291 are lower, as expected. For stomach, banding on stomach dose rather than colon dose gives 0.256 against 0.253, with no detected subgroup separation under either definition. Thus changing the organ-dose categories does not resolve the limited information about stomach effect modification.

1.10 An exploratory low-dose city association

1.10.1 The association

The overall 5–< 100 mGy contrast is compatible with zero: the exposed-group ratio-of-rates ERR is −0.019 (−0.060, 0.023) and the population-average version is −0.005 (−0.046, 0.037). Nevertheless, city modifies this contrast, and it is the only dimension that does. Of the 49 separation tests run in this fit, five have intervals excluding zero on the ratio-of-rates scale, and all five involve city.

Everything in this section is reported on the ratio-of-rates scale fixed in Section 1.6, and both averaging populations are stated once here so that no number below is ambiguous. Over the exposed cells within each city the ratio-of-rates ERR is −0.043 (−0.089, 0.003) in Hiroshima and +0.088 (0.006, 0.171) in Nagasaki, a difference of 0.132 (0.042, 0.224). Over all cells in each city, which is the summary scale the separation test uses, the values are −0.041 (−0.087, 0.005) and +0.085

(0.003, 0.167), giving differences from the cohort-averaged ERR of −0.036 (−0.063, −0.011) and +0.090 (0.026, 0.159). The two averaging populations agree to within 0.005 throughout.

The person-year-weighted mean of cell ERRs gives −0.026 (−0.101, 0.059) in Hiroshima and +0.209 (0.077, 0.365) in Nagasaki. These averages of ratios differ from ratios of averaged rates and answer a different question. Within Nagasaki, the ratio-of-rates ERR is 0.166 (0.054, 0.278) in women and 0.165 (0.022, 0.324) among those exposed at ages 30–39. These are exploratory subgroup summaries. The refitted Preston specification includes city in the baseline only; allowing a city-dependent radiation effect is an available extension of conventional LSS models, as in Grant et al. [2017].

1.10.2 Is it Monte-Carlo noise, and how large is it in dose terms?

Two questions have to be settled before the association is interpreted at all.

The first is Monte Carlo error. An independent two-chain diagnostic refit retains draws of the city summaries and estimates the Monte Carlo standard error (MCSE) of each interval endpoint separately. Nagasaki's exposed-group band ERR is 0.0863 (0.0002, 0.1739), with mean MCSE 0.00093 and lower-endpoint MCSE 0.00393. Its lower bound is only 0.06 endpoint MCSE above zero, so the apparent exclusion of zero is not numerically stable. With averaging over both exposure groups, its lower bound is −0.0030. The city difference is 0.1294 (0.0382, 0.2227), with lower-endpoint MCSE 0.00213; its positive lower bound is better resolved. These diagnostics concern numerical precision conditional on the model, not protection against confounding or model error. The main-fit subgroup summaries above and the sensitivity fits below come from separate posterior runs; their small differences near zero reinforce the need for caution about Nagasaki's own exclusion of the null.

The second is magnitude. The low-dose group's mean dose contrast is 0.0299 Gy overall, 0.0315 in Hiroshima and 0.0246 in Nagasaki. Rescaling Nagasaki's ratio-of-rates ERR by its own dose contrast gives a descriptive 3.51 (0.01, 7.07) per Gy in the diagnostic refit, and 8.54 (3.03, 14.99) per Gy for the person-year-weighted mean cell ERR. The cohort's own high-dose descriptive rescaling is 0.529 (0.419, 0.630) per Gy. These descriptive rescalings are much larger than the high-dose rescaling, showing how a small dose denominator amplifies differences in baseline risk. They do not estimate a continuous dose-response slope. The wide high-dose city-difference interval provides no evidence that the two cities are equal at high dose.

The E-value sensitivity analysis says the same thing in the other units (Table 15). On the ratio-of-rates scale an unmeasured confounder with associations of 1.39 with dose-group membership and with cancer incidence could explain Nagasaki's low-dose-group point estimate away, and 1.02 would move the interval to the null. For comparison, Table 6 gives 1.41 and 1.23 for the medium-dose all-solid result, which Section 1.5 calls less robust, and 2.34 and 2.12 for the high-dose result. The city finding therefore sits at or below the medium-dose benchmark on both measures. These E-values apply to the within-city exposed-versus-reference rate ratio. A difference of city ERRs is not a rate ratio and has no E-value under this calculation.

1.10.3 Is it new relative to the canonical model?

Section 1.7 compares the CountBCF surface with the refitted canonical risk model at the same person-year cells, and its low-dose block answers this directly. Six of the 28 low-dose subgroups depart from the canonical surface on the ratio-of-rates scale. Hiroshima is one of them, at −0.063 (−0.108, −0.016). Nagasaki is not, at +0.072 (−0.011, 0.154). On the scale this study uses for every other claim, Nagasaki's low-dose-group ERR is not distinguishable from what the canonical

Table 15: E-values and per-Gy rescaling for the low-dose city contrast, on both summary scales, from the independent diagnostic refit. Rate ratios are exposed-group and person-year weighted. The per-Gy column divides by each city's own mean dose contrast and is shown to make the size of the implied dose effect explicit, not because a per-Gy slope is estimated. For city differences, neither a within-city dose denominator nor a rate ratio is defined, so the rescaling and E-value entries are not applicable.

Summary	Quantity	Dose contrast (Gy)	ERR (95% CrI)	Implied ERR per Gy	E-value, point / interval
Ratio of rates	Hiroshima	0.0315	−0.043 (−0.089, 0.001)	−1.37 (−2.83, 0.05)	1.26 / 1.00
	Nagasaki	0.0246	+0.086 (0.0002, 0.174)	+3.51 (0.01, 7.07)	1.39 / 1.02
	Nagasaki − Hiroshima	Not applicable	+0.129 (0.038, 0.223)	Not applicable	Not applicable
Per-cell mean	Hiroshima	0.0315	−0.027 (−0.100, 0.057)	−0.85 (−3.17, 1.80)	1.20 / 1.00
	Nagasaki	0.0246	+0.210 (0.074, 0.369)	+8.54 (3.03, 14.99)	1.71 / 1.36
	Nagasaki − Hiroshima	Not applicable	+0.237 (0.093, 0.395)	Not applicable	Not applicable

Benchmarks from Table 6: all-solid medium dose 1.41 / 1.23; all-solid high dose 2.34 / 2.12.

model already predicts; Hiroshima's is. All six low-dose departures are downward, so what the estimator is mainly saying about the low-dose group is that the linear extrapolation from the whole 0–2 Gy range overshoots there, and that it overshoots most in Hiroshima. Read that way, the city contrast is at least as much a statement about Hiroshima sitting below the canonical prediction as about Nagasaki sitting above it. This motivates a reference-group sensitivity analysis, without attributing the entire city contrast to that group.

1.10.4 Sensitivity to estimator choices

The association persists across the examined modelling choices (Table 16), covering five moderating-forest tree counts, three moderating-forest depth priors, two multipliers of the leaf concentration shared by both forests, four propensity specifications, a trim, and the not-in-city group. Across all sixteen the Nagasaki-minus-Hiroshima ratio-of-rates difference runs from 0.099 to 0.146 and every 95% credible interval excludes zero, including under the tightest examined moderating-forest prior. As in the high-dose sweep, the magnitude scales with prior looseness while interval exclusion persists in these fits: the smallest value, 0.099 (0.028, 0.170), comes from halving the leaf concentration, and the largest, 0.146 (0.049, 0.245), from the most prognostic-like depth prior.

The leaf concentration a_0 is shared by the prognostic and moderating forests, whereas the tree-count and depth-prior settings vary the moderating forest alone. The trimming analysis removes the outer 5% at each end of the observed propensity-score distribution. Fixed $[0.10, 0.90]$ limits remove no cells in this low-dose comparison, whose scores span approximately $[0.50, 0.73]$, and would not test sensitivity to removing observations.

A separately fitted Poisson model with dose-group by city interactions gives the same localisation. The interaction term improves the fit at $p = 0.012$ on a likelihood-ratio test with three degrees of freedom, and the improvement is carried entirely by the low-dose group (Table 17), where the rate-ratio ratio is 1.154 (1.054, 1.264). This comparison shows that the association also appears under a conventional model. The test was selected after examining the forest result and uses the same data and reference group, so it supplies neither independent replication nor protection against multiplicity or residual confounding.

Table 16: Robustness of the low-dose (5–< 100 mGy) city difference to modelling choices. Each row is a separate refit on the ratio-of-rates scale. m_τ and the depth prior act on the moderating forest only; the leaf concentration a_0 is shared by both forests. The Nagasaki-minus-Hiroshima difference excludes zero in all sixteen specifications, and also on the per-cell summary scale, which is in the accompanying CSV. These fits use the shorter sensitivity MCMC budget of one chain with 1,000 burn-in and 1,000 retained draws, so the default row reads 0.128 against the 0.132 obtained from the two-chain headline fit.

Specification	Cells	Hiroshima group ERR	Nagasaki group ERR	Nagasaki − Hiroshima (95% CrI)
$m_\tau = 10$ (tightest)	8,541	−0.033	0.079	0.112 (0.025, 0.200)
$m_\tau = 25$	8,541	−0.034	0.082	0.116 (0.026, 0.207)
$m_\tau = 50$ (default)	8,541	−0.035	0.093	0.128 (0.038, 0.217)
$m_\tau = 100$	8,541	−0.042	0.101	0.143 (0.051, 0.238)
$m_\tau = 200$ (loosest)	8,541	−0.040	0.103	0.142 (0.050, 0.241)
Depth prior (0.05, 4), tight	8,541	−0.040	0.083	0.123 (0.036, 0.217)
Depth prior (0.5, 2), loose	8,541	−0.042	0.102	0.144 (0.051, 0.243)
Depth prior (0.95, 2), prognostic-like	8,541	−0.043	0.103	0.146 (0.049, 0.245)
$a_0 \times 0.5$ (both forests)	8,541	−0.047	0.052	0.099 (0.028, 0.170)
$a_0 \times 2$ (both forests)	8,541	−0.044	0.088	0.132 (0.039, 0.220)
Propensity score constant	8,541	−0.038	0.089	0.127 (0.034, 0.224)
Propensity score simple logistic	8,541	−0.040	0.096	0.136 (0.044, 0.228)
Propensity score rich logistic	8,541	−0.038	0.087	0.126 (0.040, 0.222)
Propensity score random forest	8,541	−0.028	0.094	0.122 (0.022, 0.227)
Trimmed to the central 90% of $\hat{\pi}$	7,687	−0.047	0.088	0.135 (0.046, 0.224)
Not in city added to reference group	9,826	−0.039	0.103	0.142 (0.050, 0.239)

Table 17: Dose-group by city interaction terms from an adjusted Poisson model. The interaction is confined to the low-dose group. Entries are rate-ratio ratios relative to Hiroshima with 95% confidence intervals.

Interaction term	Rate-ratio ratio (95% CI)	p
5–< 100 mGy × Nagasaki	1.154 (1.054, 1.264)	0.002
100–< 500 mGy × Nagasaki	1.111 (0.976, 1.263)	0.11
500–< 2000 mGy × Nagasaki	1.001 (0.870, 1.151)	0.99

1.10.5 Reconciliation with the published city analyses

The canonical 1958–1998 analysis reported a significant low-dose dose response but did not foreground a city-specific split [Preston et al., 2007], and the 1958–2009 update found little evidence that city modified the aggregate solid-cancer response, estimating Nagasaki's radiation effect at about 12% below Hiroshima's with an interval covering no difference [Grant et al., 2017]. Those analyses are not in conflict with what we report. They evaluate city differences in the magnitude and shape of the dose response across the full dose range, and they were motivated principally by dosimetry, in particular the larger neutron contribution in Hiroshima. A full-range comparison is dominated by the high-dose contribution, where our city-difference interval is wide and includes zero. What is genuinely under-examined in the published literature is a city comparison confined to the lowest group, and that is the gap this section occupies.

1.10.6 Reference-group composition

Reference-group composition is one plausible contributor. French et al. [2017] show that choices among zero-dose reference populations can change estimated ERRs and curvature, and examine adjustment for city-specific risk differences among distal survivors. Cologne et al. [2019] discuss how heterogeneity in background incidence affects the aggregate solid-cancer dose response. Our distance-based analyses assess whether this issue can account for the observed contrast.

Geographic and cohort-construction differences provide reasons to examine reference-group comparability across cities. Hiroshima's hypocentre was directly above the downtown area, and because the surrounding terrain is relatively flat the damage pattern was largely radial; moving outward from the hypocentre the population becomes progressively more rural in every direction, so Hiroshima's distal survivors beyond 3 km were largely rural at the time of the bombing. Nagasaki's hypocentre was north of downtown, in the Urakami district, and the Urakami valley is enclosed by mountain ranges; most of Nagasaki's distal survivors, 3 km and beyond to the south and southeast, were urban residents. The same nominal stratum, "zero-dose survivor beyond 3 km", therefore denotes a rural population in one city and an urban one in the other. A further asymmetry comes from cohort construction: after the initial selection of LSS participants, additional distal survivors were added to the Nagasaki cohort, randomly selected to match the original Nagasaki cohort on sex and age, which raises both the number and the proportion of distal survivors in Nagasaki. We note that "distal" is conventionally ≥ 3 km although the original selection used ≥ 2.5 km, because dose estimates were not available when the cohort was assembled and distance was the best available surrogate for separating exposed from virtually unexposed survivors.

Our own reference group has exactly this structure, and the imbalance is large (Table 18). In Hiroshima 73.9% of reference-arm person-years come from beyond 3 km; in Nagasaki, 55.7%. Meanwhile the distal ring carries a higher adjusted solid-cancer rate than the outer proximal ring in both cities, which is the direction French et al. [2017] report. Pooling the two rings therefore raises the reference rate more in Hiroshima than in Nagasaki. A higher reference rate lowers the estimated ERR in both cities, with a larger reduction in Hiroshima. This produces a city contrast in the observed direction, but does not account for its full magnitude.

A simple calculation puts the possible contribution in perspective. Weighting the two adjusted ring rates by each city's own person-year shares gives a pooled reference rate of 63.3 in Hiroshima against 60.9 for the outer proximal ring alone, a ratio of 1.040, and 57.5 against 56.5 in Nagasaki, a ratio of 1.017. Moving from the pooled reference to the outer proximal reference should therefore move the two cities' ERRs apart by roughly $1.040/1.017 - 1 = 0.023$ in Hiroshima's favour, that is, shrink the Nagasaki-minus-Hiroshima difference by about 0.023. The observed shrinkage is 0.029,

Table 18: Composition of the 0–< 5 mGy reference group by city and distance ring, with the 5–< 100 mGy group for comparison. Crude rates are cases per 10^4 person-years; the adjusted rates come from a Poisson model with the same sex-specific spline baseline used throughout. The exposed group contains no cell beyond 2.71 km, so the distal ring has probability zero of low-dose-group membership.

City	Stratum	Ground distance (km)	Subjects	Person-years	Crude rate	Adjusted rate
Hiroshima	Reference, outer proximal	2.47–3.00	5,197	139,394	64.7	60.9
	Reference, distal	3.32–7.58	14,842	394,595	67.0	64.2
	Low group, 5–< 100 mGy	1.60–2.51	21,105	566,405	61.5	N/A
Nagasaki	Reference, outer proximal	2.52–2.97	6,836	170,049	52.1	56.5
	Reference, distal	3.22–6.97	8,670	214,161	54.8	58.2
	Low group, 5–< 100 mGy	1.63–2.71	6,684	163,199	56.4	N/A

about one fifth of the baseline contrast of 0.134; the predicted 0.023 is about one sixth of it.

Table 19 reports what actually happens. Two constructions are the ones the literature recommends: restricting the reference group to the outer proximal ring, which is the RERF practice, and keeping the full reference group but giving the prognostic forest a distal indicator, which is the French et al. [2017] recommendation of adjusting for city-specific risk differences among distal survivors. Under both, the Nagasaki-minus-Hiroshima difference falls from 0.134 (0.041, 0.231) to 0.105 (−0.018, 0.222) and 0.106 (−0.011, 0.223) respectively, and in both the interval covers zero. This loss of interval exclusion is not evidence that the association has been fully explained. With the distal indicator in the prognostic forest, Hiroshima moves from −0.042 to +0.005, whereas Nagasaki moves from 0.092 to 0.111 (0.015, 0.211), remaining above zero. The city-by-distance constructions show the same broad pattern. The reference adjustment primarily accounts for Hiroshima's deficit; a modest Nagasaki excess remains unexplained. The same attenuation occurs for the cell-average ERR, where the difference falls from 0.225 (0.090, 0.388) to 0.154 (−0.020, 0.324) and 0.145 (−0.037, 0.322).

Two qualifications on that result. First, the effect is a 22% attenuation of the point estimate combined with a wider interval, and the widening is partly mechanical: restricting the reference to the outer-proximal ring reduces total analytic cases from 10,009 to 6,194, so precision is lost as well as signal. What the exercise establishes is not that the difference is zero but that it is not robust to the reference-group choice known to matter most for exactly this question. Second, adding the ring to the propensity score rather than to the prognostic forest attenuates less, to 0.122 (0.009, 0.239) and 0.118 (0.015, 0.233), and those intervals stay marginally clear of zero; but those are also the specifications in which the fitted score runs down to 5×10^{-9}, because the distal reference cells have probability exactly zero of low-dose-group membership. Conditioning on the ring in the score is the wrong instrument for that reason, which is precisely why the recommended constructions handle the distal group through the baseline instead.

Conditioning on continuous ground distance in both the score and prognostic forest reverses the city difference to −0.084 (−0.332, 0.156), while the fitted propensity spans 2×10^{-16} to 1.000. The lack of overlap makes this fit unsuitable for a causal interpretation; it documents how strongly the estimate depends on extrapolation when the dose-assignment variable is included.

The parametric model also shows attenuation of the city association (Table 20). Refitting the dose-group by city interaction of Table 17 under the two recommended reference constructions moves the joint likelihood-ratio test from $p = 0.012$ to $p = 0.106$ and $p = 0.108$, and the low-dose-group interaction from 1.154 (1.054, 1.264) at $p = 0.002$ to 1.131 (1.006, 1.273) at $p = 0.040$ and 1.131 (1.005, 1.272) at $p = 0.041$. Restricting the reference to the distal ring instead, leaves the

Table 19: The low-dose city difference under alternative treatments of the $< 3/\geq 3$ km distance ring. All entries are on the ratio-of-rates scale over exposed cells; the per-cell column is given for cross-reference. The first two rows after the baseline are the constructions recommended by RERF practice and by French et al. [2017]. The last row conditions on the continuous assignment variable and is reported only to document the sign reversal and the positivity collapse.

Specification	Cases	$\hat{\pi}$ range	Nagasaki − Hiroshima, ratio of rates (95% CrI)	Same, per-cell (95% CrI)	Separates?
Baseline, full 0–< 5 mGy reference	10,009	0.50–0.73	0.134 (0.041, 0.231)	0.225 (0.090, 0.388)	Yes
Reference = outer proximal only (2.5–3.0 km)	6,194	0.69–0.84	0.105 (−0.018, 0.222)	0.154 (−0.020, 0.324)	No
Distal indicator in the prognostic forest	10,009	0.50–0.73	0.106 (−0.011, 0.223)	0.145 (−0.037, 0.322)	No
Reference = distal only (3–10 km)	8,221	0.61–0.85	0.103 (0.014, 0.196)	0.170 (0.049, 0.308)	Yes
City × distance in the score only	10,009	5×10^{-9}–0.84	0.118 (0.015, 0.233)	0.154 (−0.008, 0.324)	Yes
City × distance in score and forest	10,009	5×10^{-9}–0.84	0.122 (0.009, 0.239)	0.182 (−0.024, 0.374)	Yes
Continuous ground distance in score and forest	10,009	2×10^{-16}–1.000	−0.084 (−0.332, 0.156)	−0.063 (−0.289, 0.151)	No

joint test below 0.05 at $p = 0.029$.

Table 20: The parametric dose-group by city interaction under four reference-group constructions. Entries are rate-ratio ratios relative to Hiroshima with 95% confidence intervals; the final column is the joint likelihood-ratio test on three degrees of freedom for the whole interaction.

Reference construction	5–< 100 mGy × Nagasaki	p	Joint LRT p
Primary, full 0–< 5 mGy reference	1.154 (1.054, 1.264)	0.002	0.012
City-specific distal baseline term	1.132 (1.006, 1.273)	0.040	0.106
Reference = outer proximal only	1.131 (1.005, 1.272)	0.041	0.108
Reference = distal only	1.151 (1.042, 1.272)	0.006	0.029

1.10.7 Dose subcategories and the unresolved shape of the association

A simple linear dose effect predicts an increasing ERR with dose, whereas a fixed population difference between the exposed and reference groups can produce a level shift. Splitting the low-dose group into 5–< 20, 20–< 50 and 50–< 100 mGy allows an exploratory comparison of these patterns.

Nagasaki's exposed-group band ERRs are 0.118 (0.010, 0.226), 0.044 (−0.077, 0.166) and 0.101 (−0.067, 0.287), respectively (Table 21). The city differences are 0.167 (0.049, 0.282), 0.111 (−0.024, 0.253) and 0.110 (−0.069, 0.298). The point estimates show no clear increasing trend, but the intervals overlap widely. Exposed cases decline from about 2,080 in the lowest subcategory to about 880 in the highest. These fits therefore do not distinguish a level shift from a radiation-related gradient. The large changes in descriptive ERR/Gy rescalings also reflect the small dose denominators; they cannot establish the absence of a dose response.

Table 21: The low-dose group split by dose, each subcategory contrasted against the same 0–< 5 mGy reference. The point estimates do not increase systematically with dose, but their wide, overlapping intervals leave the dose-response shape unresolved. The per-Gy values are descriptive rescalings.

Sub-group	Dose contrast (Gy)	Hiroshima group ERR	Nagasaki group ERR	Nagasaki − Hiroshima	Sep.?	Nagasaki ERR/Gy
5–< 20 mGy	0.0096	−0.049 (−0.109, 0.011)	+0.118 (0.010, 0.226)	0.167 (0.049, 0.282)	Yes	13.0
20–< 50 mGy	0.0353	−0.068 (−0.129, −0.002)	+0.044 (−0.077, 0.166)	0.111 (−0.024, 0.253)	No	1.28
50–< 100 mGy	0.0743	−0.010 (−0.087, 0.069)	+0.101 (−0.067, 0.287)	0.110 (−0.069, 0.298)	No	1.37

1.10.8 Explanations the data address less well

The contrast persists after cancer-site exclusions. Repeating the low-dose-group contrast on the aggregate with breast and thyroid removed, with non-melanoma skin removed, with stomach removed, and restricted to non-sex-specific sites gives a Nagasaki-minus-Hiroshima difference of 0.129 (0.031, 0.228), 0.124 (0.032, 0.222), 0.142 (0.034, 0.251) and 0.107 (0.016, 0.206) respectively, against the baseline 0.134 (0.041, 0.231); all four still separate. The result is not confined to the excluded sites. In contrast, Sasaki et al. [2014] attribute a low-dose Nagasaki excess mainly to lung, liver and gallbladder. The persistence across site exclusions suggests that the association is not confined to one of these cancer groups, while leaving its explanation open.

Dose assignment remains a live possibility that we cannot settle. The two cities have different shielding histories and different gamma-to-neutron mixtures, so an error in the neutron weighting or in the distal dose imputation would present as a city difference in risk per weighted Gy. The neutron fraction is small at these doses, but the uncertain high-dose city comparison cannot rule out a dosimetric explanation. Residual and fallout dose is absent from DS02 and could differ by city. The public file cannot resolve these possibilities.

Case ascertainment cannot be evaluated directly from this table. LSS incidence, unlike mortality, depends on the two tumour registry catchment areas, and person-years are adjusted for migration into and out of them [Grant et al., 2017]; that adjustment is city-specific and varies with distance from the city centre, which is the axis of this finding. Comparability of migration adjustment, case ascertainment, registry quality, Adult Health Study participation and examination protocols requires information beyond this public table. The Nagasaki distal augmentation described in Section 1.10.6 is one documented asymmetry in the reference-group composition. A useful check would be to repeat the low-dose-group city contrast using LSS mortality data, which have different ascertainment requirements. This analysis is left for follow-up (Section 1.15).

Multiplicity cannot be dismissed either. This is one selected result among 294 separation tests (Table 8), and $p = 0.012$ from the parametric interaction is not overwhelming, particularly since that test was run on the same data after the forest result was known.

1.10.9 What we conclude

The low-dose city contrast persists across the modeling choices examined and is also visible in a conventional Poisson interaction fit on the same data. It remains exploratory. Reference-group adjustments attenuate the contrast by about one fifth and account for Hiroshima's deficit, but leave a modest Nagasaki excess unexplained. The separate dose-category fits provide no clear increasing gradient within the low-dose range, and cannot exclude one given their precision. Residual confounding, dose error, ascertainment, and multiplicity remain possible explanations. These analyses illustrate how a flexible effect surface can identify comparisons for further epidemiological investigation without establishing a new city difference in radiosensitivity.

1.11 Propensity score and overlap

The primary propensity score is a logistic regression of group membership on city by sex, natural splines with four degrees of freedom in age at exposure and in attained age, and a city by age-at-exposure interaction. Ground distance is deliberately excluded because it is the assignment mechanism rather than a confounder. Four alternatives were run for comparison: a constant score, a plain additive logistic model, a saturated spline model, and a probability random forest.

Demographic overlap is adequate in the primary fits. For the featured high-dose contrast the primary score spans 0.117 to 0.752, with no cells below 0.05 or above 0.95, and the same holds for

the low and medium contrasts and for every logistic specification. The random forest has an out-of-bag area under the curve of 0.77, compared with about 0.56 for the logistic models, and pushes two cells above 0.95. Its out-of-bag predictions measure held-out discrimination; that AUC alone is not evidence of overfitting. The score is nearly flat because dose was determined by location and shielding rather than by demography, so the discrimination of the logistic model is barely above chance. Figure 8 displays the distributions.

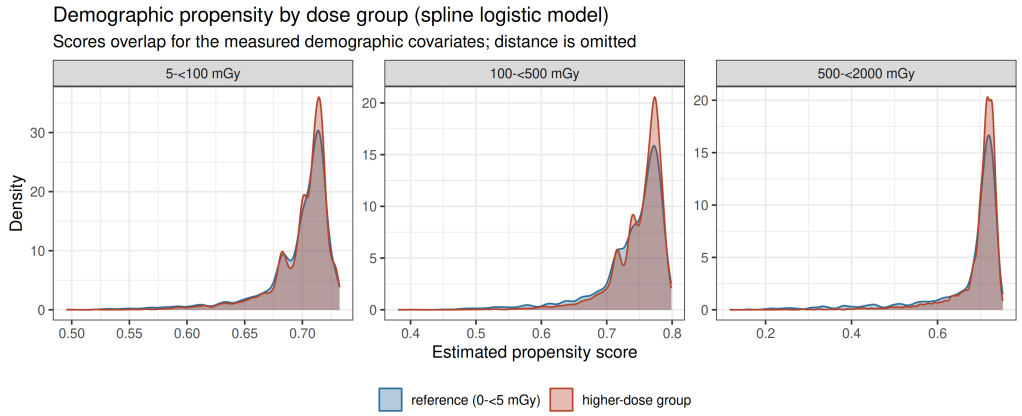


Figure 8: Propensity-score distributions by dose group for each dose-group contrast. No cell falls below 0.05 or above 0.95 under the primary specification.

Weighting removes what imbalance is present. Cell-level standardized mean differences before weighting run from 0.056 to 0.110 for city and reach −0.206 for age at exposure; after inverse-probability weighting with the primary score they are −0.003 to 0.002 for city, −0.002 for age at exposure, −0.003 for attained age, and −0.002 for sex. Trimming is a no-op: because no cell falls outside $[0.10, 0.90]$, trimming at either $[0.05, 0.95]$ or $[0.10, 0.90]$ removes nothing and returns numerically identical results, with all three variants giving 8,249 cells and an exposed-group ratio-of-rates ERR of 0.495. The trims therefore provide no empirical sensitivity check for this contrast. These diagnostics concern overlap in the measured demographic covariates; they do not resolve the lack of overlap by distance.

Table 22: Sensitivity of the featured contrast to the propensity-score specification. Even a constant score reproduces the primary estimate, because the prognostic forest already conditions on city, sex, and both age variables.

Propensity specification	Cells	Ratio-of-rates ERR	EAR/10^4 PY (95% CrI)	Mean-cell ERR, < 20 minus 40+
Constant (no covariates)	8,249	0.497	30.8 (25.1, 36.7)	0.815 (0.473, 1.19)
Simple logistic	8,249	0.500	30.9 (25.1, 36.9)	0.844 (0.494, 1.21)
Spline logistic (primary)	8,249	0.495	30.7 (25.3, 36.5)	0.795 (0.447, 1.14)
Saturated spline logistic	8,249	0.491	30.6 (25.4, 36.2)	0.773 (0.429, 1.13)
Probability random forest	8,249	0.497	30.8 (25.3, 36.4)	0.913 (0.464, 1.39)

The estimates change little across these score specifications (Table 22). Even a constant propensity gives 0.497 against the primary 0.495, because the prognostic forest is already conditioning on city, sex, and both ages, leaving the score little to correct. This demonstrates limited sensitivity to these score choices for the measured demographic covariates, while exchangeability remains an assumption.

The featured subgroups are also not in a low-overlap tail. Within the exposed arm of the high-dose contrast, the youngest-at-exposure cells, which carry the large ERR, have the tightest propensity distributions of all, spanning 0.700 to 0.731 for males exposed at ages 0–9 and 0.714 to 0.742 for females, with no cell below 0.05 or above 0.95. The only subgroup with any low-propensity cells is those exposed at age 40 or later, with a minimum of 0.183, and that subgroup sits at the low end of the ERR gradient. The prominent age gradient does not concentrate in cells with extreme values of this demographic propensity score.

One limitation of these diagnostics has to be stated explicitly, because it is the hinge of Section 1.10. They are computed from a propensity model that deliberately omits ground distance, the assignment variable. What they establish is that the arms overlap on the demographic covariates, which they do, comfortably. What they cannot establish is overlap on anything distance-correlated, because within a city the dose group is very nearly a distance ring: the 5–< 100 mGy group runs from about 1.6 to 2.7 km from the hypocentre and the reference group from 2.5 km outward, with no demographic variable in the file distinguishing the two. A flat propensity score and clean balance tables are consistent with a large unadjusted difference in geographic composition between the two rings. The reference-group comparisons in Section 1.10 identify baseline-rate variation along this distance axis, without measuring urban and rural composition directly. The sharpest statement of the problem is that the 5–< 100 mGy group contains no cell beyond 2.71 km, while 74% of Hiroshima's reference person-years and 56% of Nagasaki's come from beyond 3.22 km. Those distal reference cells have a probability of low-dose-group membership of exactly zero, which is the same positivity argument that Section 1.2 uses to exclude the not-in-city group, applied one ring closer in. Adding the ring to the propensity model makes this visible immediately: the fitted score then runs from essentially zero rather than from 0.50.

1.12 Repeated person-years

Each survivor contributes person-years to many cells as they move through attained-age and calendar-period categories, so an independent-cell likelihood can understate uncertainty if within-cohort dependence remains. We quantify this with a frequentist cluster bootstrap of the parametric model, resampling the 1,128 survivor cohorts, defined by city, sex, age at exposure, dose category, and distance category, with replacement over 300 replicates (Table 23).

Table 23: Cluster bootstrap of the parametric model over 300 replicates of the 1,128 survivor cohorts. The design effect is the ratio of the bootstrap sampling variance to the model variance (the square of the ratio of the bootstrap standard error to the model standard error).

Dose group	Coefficient	Model SE	Cluster-bootstrap SE	Design effect
Low	−0.0036	0.0205	0.0211	1.06
Medium	0.1112	0.0259	0.0247	0.91
High	0.4261	0.0325	0.0346	1.14

For this parametric model, the largest bootstrap standard error is about 7% above the model-based standard error. This is a sensitivity check for that model, not a bound on uncertainty in CountBCF's posterior intervals [Field and Welsh, 2007, Kish, 1995].

The available random-effects arguments do not add a cohort effect to the count likelihood, so no CountBCF random-intercept analysis is reported. A separate mean-model sensitivity analysis supplies a cohort identifier as one integer predictor in the prognostic forest. Using a 60-level city-by-sex-by-exposure-age code gives an exposed-group ERR of 0.494 (0.399, 0.592), compared

with 0.495 (0.401, 0.594) without the code; the 432-level finest-cohort code gives 0.507 (0.403, 0.616). These identifiers can change the fitted mean, with splits constrained by the arbitrary integer ordering. They do not model within-cohort correlation or correct credible intervals for clustering. Valid inference under such dependence requires a likelihood or resampling scheme that explicitly represents it.

1.13 Low-dose sub-cohorts

Dropkin [2016] re-analysed this same file with generalized additive models restricted to women and to the 0–20, 0–100, and 0–500 mGy sub-cohorts, and reported relative risks at 10 mGy that are ten to forty-five times higher than a linear extrapolation from the full cohort. The same restrictions were run here (Table 24).

Table 24: CountBCF fits on low-dose sub-cohorts. The dose contrast column is the mean weighted-dose separation between the exposed and reference groups.

Sub-cohort	Cells	Cases	Dose contrast (Gy)	Ratio-of-rates ERR (95% CrI)	EAR/10^4 PY (95% CrI)
5–< 100 mGy, women	4,360	5,474	0.029	0.0006 (−0.061, 0.059)	0.01 (−3.43, 3.13)
5–< 100 mGy, men	4,181	4,535	0.031	−0.047 (−0.110, 0.012)	−3.60 (−8.61, 0.90)
5–< 20 mGy, both sexes	3,839	7,687	0.0096	−0.005 (−0.057, 0.049)	−0.28 (−3.36, 2.90)
5–< 20 mGy, women	1,938	4,239	0.0096	−0.004 (−0.075, 0.065)	−0.25 (−4.09, 3.47)

CountBCF finds no low-dose excess in any of them. This is not a refutation of Dropkin [2016] and should not be presented as one, because the two designs answer different questions. His model fits a smooth dose response in a transformed dose coordinate, the empirical cumulative distribution function of dose, and reads a risk off it at 10 mGy, borrowing strength across the whole low-dose range. Ours contrasts a 5–< 20 mGy group against a 0–< 5 mGy group, a mean dose separation of only 9.6 mGy, with no functional form to borrow from. At that separation, the design has limited precision for per-Gy inference: the credible interval on the ERR, at roughly ±0.07, corresponds to ±7 per Gy, which cannot resolve the small low-dose effects of primary epidemiological interest. A continuous-dose extension would allow a closer comparison with that point-dose analysis, whereas the present model estimates category contrasts.

1.14 Excess cases and attributable fractions

Converting the exposed-group effect back to a case count gives the comparison in Table 25. At medium and high dose the two approaches agree closely, at 372 against 402 excess solid cancers in the 0.5–2 Gy range and 178 against 254 in the 0.1–0.5 Gy range. The gap widens at low dose, where CountBCF returns an interval containing zero and a slightly negative point estimate while the parametric model assigns 81 excess cases. That is the linear dose-response assumption doing work: the parametric model borrows the slope from the whole 0–2 Gy range and applies it at 30 mGy, whereas the group contrast must find the signal within the group itself. Neither is wrong, but they answer different questions, and the difference illustrates concretely what a nonparametric causal estimator will and will not deliver.

1.15 Limitations

The causal interpretation requires exchangeability, consistency, no interference, and adequate overlap. These conditions are not established by the fit. Dose is closely tied to distance, and demographic overlap does not remove possible distance-related confounding. Reference-group adjust-

Table 25: Excess cases and attributable fractions from CountBCF, compared with the corresponding published estimates. AF denotes the attributable fraction among the exposed.

Endpoint	Dose group	Observed cases	CountBCF excess (95% CrI)	CountBCF AF	Published excess	Published AF
All solid	5–< 100 mGy	4,406	−86 (−274, 103)	−2.0%	81	1.8%
	100–< 500 mGy	2,112	178 (68, 286)	8.4%	254	12.0%
	500–< 2000 mGy	1,148	372 (300, 438)	32.4%	402	35.0%
Stomach	5–< 100 mGy	1,230	−58 (−165, 48)	−4.7%	14	1.2%
	100–< 500 mGy	555	0.3 (−57, 60)	0.0%	46	8.2%
	500–< 2000 mGy	269	54 (21, 89)	20.0%	69	25.9%

ments explain Hiroshima's deficit and about one fifth of the exploratory city contrast; Nagasaki retains a modest unexplained excess. The high-dose city interval is too wide to establish equality or rule out a city-dependent risk pattern.

The summaries compare dose groups over specified covariate populations. Dividing an ERR by the mean dose contrast is descriptive and does not produce a fixed-profile continuous-dose coefficient. At low dose the small denominator amplifies modest differences in background risk. Neither the broad dose categories nor their intervals resolve the low-dose dose-response shape.

The high-dose sex contrast is sensitive in magnitude to regularisation and loses interval exclusion under several site-composition changes, including removal of breast and thyroid. The flat stomach surface is absence of detected modification in a smaller dataset, not proof of homogeneity. Subgroup analyses are exploratory, use overlapping comparisons, and are not multiplicity adjusted. The parametric city test uses the same data and is not independent replication.

Cells aggregate survivor follow-up and contain no individual smoking, diet, or infection measures. The parametric cluster bootstrap does not calibrate CountBCF intervals, and adding an integer cohort code to the forest adjusts only the mean. Credible intervals are conditional on the fitted independent-cell likelihood, prior, and estimated propensity scores. The simulation settings have smaller samples, no person-year offsets, much lower dispersion, and known propensities; their coverage is not direct evidence for applied interval coverage. The fixed-age profile calculation retains between-chain disagreement and does not yet support a stable comparison with published sex-specific risks.

Registry ascertainment, migration adjustment, and unmeasured background-risk factors cannot be evaluated fully from the public table. DS02 doses include the released dose-error adjustment, but further measurement error and residual or fallout dose are not modeled. Analyses using additional individual covariates or alternative follow-up outcomes would help distinguish these explanations.

1.16 Summary

On the public LSS incidence table, CountBCF recovers the broad all-solid-cancer risk pattern and the principal age-at-exposure and attained-age gradients without imposing their functional forms. The exposed-group high-dose ERR is 0.487 (0.386, 0.580) for 0.5–2 Gy versus < 5 mGy. Its mean-dose rescaling is close to published per-Gy summaries, while differences in covariate profile, follow-up, and uncertainty definitions limit direct numerical comparison. The age-at-exposure gradient persists across the sensitivity analyses; sex differences are more sensitive to prior magnitude and cancer-site composition. Stomach data provide little evidence about effect modification.

The exploratory low-dose city contrast illustrates both the method's use and its limits. The composition of the reference group accounts for Hiroshima's deficit and about one fifth of the contrast, leaving a modest Nagasaki excess unexplained. The broad high-dose interval, limited low-dose information, unmeasured covariates, and dependent follow-up prevent a firm etiologic

interpretation. CountBCF offers a flexible complement to conventional LSS models and identifies comparisons that warrant further study.

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